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Back cover designed by all the 5D Project groups featured in this issue.

Additional Information

This issue of *SupBiotech Projects* contains research articles written by students enrolled in the Apprenticeship engineering program at SupBiotech, as part of their contribution to the Undergraduate Research Program in Bioengineering (5D Projects), supervised by Julien Grimaud, PhD. For more information, please contact Dr. Grimaud using the following address: julien.grimaud[at]supbiotech.fr

Editorial

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Welcome to Volume 7 of *SupBiotech Projects*! This edition marks another milestone in our continuous journey of exploration and innovation in the realm of biotechnology. As we delve into the pages of this journal, we are presented with a compelling collection of research papers authored by the brilliant minds of the graduating Apprenticeship class of 2024.

The articles presented in this volume represent the culmination of the 5D Projects, a distinctive component of SupBiotech's Undergraduate Research Program in Bioengineering. These transversal research and development projects accompany our students throughout their academic journey at SupBiotech, fostering growth, creativity, and teamwork. The 5D projects revolve around the 5D methodology – Discover, Define, Design, Develop, Deploy.

Each project unfolds across four distinct phases, guiding students through the intricacies of biotechnological exploration. The journey begins with groups conducting scientific reviews on topics of their choice, paving the way for innovative ideas and solutions. Subsequently, they delve into the realm of economic viability, meticulously navigating market landscapes to refine their concepts. The next stage brings them into SupBiotech Teaching Laboratories, where they harness their theoretical knowledge to collect preliminary scientific data, showcasing their prowess in experimental design and execution. The final step culminates in the creation of a comprehensive scientific article.

To uphold the highest standards of quality, each paper has undergone rigorous peer-review by groups of SupBiotech students. As you peruse through this journal, we trust it will

serve as a valuable resource for current and future students, as well as researchers across diverse fields of biotechnology.

Our heartfelt gratitude extends to everyone who contributed to the creation of this journal, from the unwavering support of the Office of Apprenticeship Student Affairs to the indispensable role played by the Teaching Laboratories team. Special thanks to Claire Grassi for her invaluable assistance during the market research phase, laboratory manager Baptiste Favard for his continuous dedication to the success of the projects, and to all members of the Research Laboratories at SupBiotech for their guidance and support.

A special commendation to the *Proryzon* group: Laurie EUVRARD, Océane FOUCHER, Allan LECONTE, Octave MAKELELE, Laurine MOENNE-LOCCOZ, Camille PAVARD, and Léa RENOARD. Instead of submitting an article to this journal, the group focused on preparing a comprehensive application to *UCook Innovation*, a student research competition focused on food innovations. *Proryzon* has made it to the finals, which will be held in Guingamp in March 2024. Fingers crossed!

Last but certainly not least, we extend our warmest thanks to the Apprenticeship class of 2024. As torchbearers of the *5D Projects*, you have not only pushed the boundaries of scientific inquiry but also played a pivotal role in refining and shaping this groundbreaking undergraduate research program. Your hard work, constructive feedback, and dedication have left an indelible mark. Embrace the pride in your accomplishments and the remarkable milestone you've achieved. Here's to your success, and may your future endeavors be as bright and promising as the discoveries within these pages!

The Pleasure of Tasting: A Hedonic Analysis of a Food Product in an Academic Context

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Abstract – This scientific article introduces the concept of hedonic tests as a method for sensory and subjective evaluation, widely applied in psychology, neurology, marketing, and food research. While commonly used in professional settings, academic applications face challenges due to resource constraints. The article presents a methodology for an accessible and cost-effective hedonic test, demonstrated through a study comparing two fermented spirulina-based food products. The objective is to assess gustatory appreciation, addressing the initial goal of enhancing plant-based protein through fermentation. The comprehensive methodology covers participant selection, sample designation, test conditions, and parameters evaluation. The study emphasizes the use of tools like Microsoft Forms, Excel, and R software for results analysis, employing statistical tests like Wilcoxon and ANOVA. The article concludes by demonstrating the feasibility of implementing a hedonic test within academic projects with limited resources, providing a valuable contribution to student research in the field.

Index Terms – Methods, Hedonic test, Statistic tests, Food processing.

INTRODUCTION

In a scientific context, a hedonic test is a method of sensory and subjective evaluation used to measure the degree of pleasure, satisfaction, or appreciation that a person experiences in response to a stimulus, typically related to organoleptic perception, such as sight, taste, smell, hearing, or touch. It is commonly used in various fields such as psychology, neurology, marketing, food research, and many others. The data collected through these tests are of crucial importance for decisions such as the launch, improvement, or commercial strategy of a product, as well as for understanding consumer behaviors. In some cases, these tests aim to assess and compare the overall appreciation of various products by focusing on individual sensations related to pleasure or

displeasure elicited, for example in the case of food products, cosmetics, etc.[1]

Despite their usefulness in the professional world, hedonic test protocols are less prevalent in academic settings, creating a challenge for students and research projects with limited financial resources. In fact, an industrial-scale hedonic test requires a lot of resources. As a result, some companies use subcontractors or experts who must be paid. In addition, the larger the panel of testers, the more usable the results, but the higher the cost of setting up the equipment. Nonetheless, hedonic tests are invaluable for identifying areas for improvement in student projects and for establishing their credibility by demonstrating that a rigorous product evaluation study has been carried out, backed up by in-depth statistical analysis of the data.

When setting up a hedonic test within the framework of an engineering school project, the difficulty of finding suitable protocols prompted us to explore the methodology presented in a study by C. Vindras and N. Sinoir describing several easy-to-implement procedures. Thus, we drew inspiration from this to set up our own method.[2] The central objective of our study is to present an accessible and usable method for carrying out a hedonic test. The test was conducted specifically between two fermented spirulina-based food products, and we will detail the entire process of implementing the test, focusing on the academic and accessible aspects.

The project originated from the initial desire to enhance plant-based protein through fermentation, producing a food product from a microalga, Spirulina. Given that spirulina is often criticized for its taste and knowing that fermentation can alter the flavor of food, the goal is to evaluate the gustatory appreciation of food products.[3] To achieve this, the study materialized through the execution of a comparative hedonic test between the two fermented products to assess consumers' preference.

Here we propose a comprehensive methodology for carrying out a hedonic test which is a forced choice test, for students, highlighting the objectives and conditions for implementing the test. It will detail the reliable, impartial, and usable tools and methods used to test the products and the use of the results obtained. Emphasis will also be placed on the objective implementation of the test while ensuring fair treatment for all concerned. Strategies for minimizing tester influence and establishing unbiased test conditions will be discussed. The limitations of the test will be pointed out. Overall, the aim is to show the feasibility of a hedonic test deployed as part of a school project with limited resources.

MATERIAL AND METHOD

I. COMPARISON HEDONIC TEST

The hedonic test, at the core of our study, was designed to assess participants' sensory satisfaction with two products we developed. This section details the rigorous methodology employed to successfully conduct this test, from participant selection to result analysis.

Participants

The study involved the participation of 35 individuals, aged between 18 and 50 years including 13 men and 22 women. Prior to the hedonic test, candidates were invited to register, facilitating test organization. The registration request was disseminated to all school students and faculty via email. Each candidate's test time was communicated individually via email, and appointments were scheduled to block a 30-minute slot in their agenda.

Samples

The hedonic test aimed to evaluate two products (YALI and TALI) we developed to determine which would be better appreciated and have greater market potential. Details on these samples are presented in a subsequent section. Each sample was designated by an alphanumeric code (5T7 and 3M8) to minimize any influence on the evaluators.

Test Conditions

Tests were conducted in groups of up to 6 participants in a classroom reserved in advance. Each participant was assigned to a specific table (6 tables) to prevent communication between participants (verbal, gestural, or visual), Appendix 1. To minimize environmental influence, the test procedure was explained beforehand in a video. This video reiterated the test objectives and specific conditions. Participants filled out a form available in either paper or electronic version via the Microsoft Forms tool.

Evaluated Parameters

The test form, Appendix 2, consists of a first part with general questions about the candidate, exploring parameters that could influence responses (the influence effect will be evaluated by an ANOVA test). Subsequently, evaluators rated each product individually in terms of appearance, taste, and

suitability as an appetizer product. These parameters will be evaluated using a Wilcoxon test. All analyses were conducted using the R software.

II. PRESENTATION OF THE TOOLS USED

Microsoft Forms (Microsoft 365)

Microsoft Forms is an online form creation tool that is part of the Microsoft 365 suite. It enables easy creation of surveys with various question types. Forms can be shared through links, email, or embedded in websites. Real-time responses are collected, and results can be analyzed using automatic charts. Integrated with Microsoft 365, Forms offers compatibility with other applications such as Excel, along with customization and security options.[4]

Microsoft Excel (Microsoft 365, version 16.80)

To analyze our results, we used both R and Excel software. Excel is a spreadsheet developed by Microsoft, primarily used for organizing, analyzing, and visualizing data in the form of worksheets. It offers features such as formulas, charts, analysis tools and graphs like the Boxplot.[5]

R software (Rstudio version 2022.02.0)

On the other hand, R software is a programming language and statistical environment. It is widely used for data analysis, statistical modeling, and graph creation. It is an open source, meaning its community can contribute to its development, making it a powerful tool for statisticians and data scientists.[6]

Wilcoxon signed-rank test

We used the Wilcoxon signed-rank test, which is a non-parametric statistical test used to compare two related samples. The non-parametric nature of this test is relevant for our study as our data does not follow a normal distribution. The hypotheses formulated to validate this test are as follows: Null Hypothesis (H0): There is no significant difference in the preference between the two products. Alternative Hypothesis (H1): There is a significant difference in the preference between the two products. The significance level for this test is 5%.

If the p-value is greater than 5%: H0 is accepted, indicating no significant difference in the preference between the two products. If the p-value is less than 5%: H0 is rejected, indicating a significant difference in the preference between the two products.[7]

ANOVA Test

The ANOVA test is a statistical test used to analyze differences between the means of three or more groups. It determines whether these differences are statistically significant. ANOVA compares intra-group variability to inter-group variability to assess whether group means are different. This test is important for demonstrating the presence or absence of bias in our hedonic test. ANOVA was used to compare the influence between tester parameters

(Age, frequency of vegetable consumption, and the table number at which they were seated) and organoleptic parameters ("Odor," "Color," "Visual texture," "Overall appearance," "Salt," "Bitterness," "Acidity," "Garlic," and "Overall taste"). The variable "Age" has been categorized into 03 values: 0, 1, and 2 corresponding respectively to the age groups 18-24 years, 25-34 years, and 35 years and above. Null Hypothesis (H0): There is no influence between tester and organoleptic parameters. Alternative Hypothesis (H1): There is an influence between tester and organoleptic parameters.

The significance level for this test is 5%. However, in our case, the R software generated error rates different from 5% based on our parameters. Some results were accepted according to error rates of 1% and 0.1%. If the p-value is greater than 5%: H0 is accepted, indicating no significant difference in the preference between the two products. If the p-value is less than 5%: H0 is rejected, indicating a significant difference in the preference between the two products. However, in our case, the R software generated error rates different from 5% based on our parameters. Some results were accepted according to error rates of 1% and 0.1%.[8]

III. SAMPLE CONCEPTION

In the context of this study, our aim was to develop innovative, healthy, and locally sourced fermented appetizer spreads, showcasing spirulina as a central ingredient. Spirulina, a microalga considered a superfood for its protein richness, was chosen as the key ingredient. The selected bacterium for fermentation belongs to the *Lactobacillus* family, a method demonstrated in a previous study.[3]

We created two distinct spreads, YALI and TALI, sharing a similar composition but differentiated by their manufacturing processes. For YALI, the fermentation of spirulina was integrated simultaneously with the other ingredients. This approach aimed to promote a synergistic interaction between spirulina and other components right from the beginning of the production process. In the case of TALI, the fermentation of spirulina was conducted beforehand, prior to the addition of other ingredients. This specific sequence aimed to explore potential variations in sensory profiles resulting from the temporal difference in the fermentation process.

RESULTS

I. HEDONIC TEST METHOD

This study focused on a hedonic test designed to assess sensory satisfaction with two developed products (YALI and TALI). Thirty-five participants aged between 18 and 50 volunteered for the experiment, and a detailed methodology, from registration to result analysis, was employed. The test evaluated appearance, taste, and suitability of the products as appetizers, using a structured form. The analysis involved statistical tools such as Microsoft Forms, Excel, R software, Wilcoxon signed-rank test, and ANOVA test. The products,

containing spirulina, were fermented differently, and the study aimed to identify preferences based on manufacturing processes. The rigorous methodology ensured a controlled and comprehensive evaluation of the products.

The flowchart in Figure 1 represents the steps to follow to carry out an accessible academic-level hedonic comparison test with the associated resources. The first column outlines the main activities to successfully conduct the hedonic test in chronological order, the second column describes tasks related to the activities. The tools used are listed in the third column, and risks and advice related to different activities are mentioned in the fourth column.

The first step is to define the scope of the test by specifying the target and the number of participants. A Forms form is useful for inviting targeted testers to the tasting. It is crucial to ensure that participants do not have any allergies or dietary restrictions for the tested products. It is essential to do this as early as possible, several weeks before the tasting day, to ensure an adequate number of participants and to send a reminder a few days before if necessary. Additionally, the format of the product evaluation must be determined whether it will be digitalized or in paper format. After that, the way the data from the tasting will be processed needs to be defined, including the choice of statistical tests. Next, define and reserve the test room, ensuring it has adequate lighting and a comfortable temperature.

At this stage, set up the entire procedure related to the smooth running of the hedonic test. A reasonably short video summarizing the rules to follow before, during, and after the tasting can be useful in ensuring all testers are under the same conditions. When the product samples are prepared uniformly and representatively, the reception should be neutral for all participants, with a random distribution of samples to avoid biases. The hedonic test begins with an evaluation of the visual appearance, followed by the smell, texture, and finally the taste of the product. Encouraging testers to take a break between different products, if necessary, by drinking water to rinse their palates is essential for obtaining actionable results. Do not forget to clean between tastings.

The test concludes with the completion of the evaluation form, followed by thanking the participants with sweets or providing feedback on the tasted product if applicable.

After that, data collection is carried out using the evaluation forms, which can then be compiled into an Excel file for analysis using the R software. Transforming qualitative values into quantitative values by assigning scores (0 and 1) (see the appendix for score correspondences) is necessary for exploiting the data through the previously chosen statistical methods.

We relied on these methods described above to conduct our hedonic test to obtain results for our two tested products based on several organoleptic parameters.

II. RESULT ANALYSIS

ANOVA Test

Table 1 presents the results of the ANOVA statistical test, conducted using the R software, focusing on the product YALI. This analysis aims to detect any significant variations between the variables "Age," "Vegetable frequency," and "Table number" concerning various quality criteria such as "Odor," "Color," "Visual texture," "Overall appearance," "Salt," "Bitterness," "Acidity," "Garlic," and "Overall taste." The R software defines error risks other than 5% during result analysis. A 1% error risk applies to values marked with a *. Hence, these results lack significance as the p-value exceeds 0.01. A 0.1% error risk applies to values marked with **. Consequently, these results lack significance as the p-value exceeds 0.001.

It is crucial to emphasize that, for a difference to be considered significant, the p-value must be less than 0.05.

Regarding the variable "Age," no statistically notable differences were observed for all the criteria. These results suggest homogeneity in the evaluations, indicating that different age groups in the hedonic test exert no significant

influence on the analyzed quality criteria. Similarly, for the variable "Vegetable frequency," no statistically remarkable variation was found for all criteria. This highlights the consistency of evaluations among vegetable consumption frequency groups, indicating no influence on quality criteria. Concerning the variable "Table number," no statistically significant fluctuation was observed for all evaluated criteria. These results suggest that the choice of the table does not significantly affect the evaluators' perceptions. Table 2 presents the ANOVA test results for the product TALI, and these results are identical to those in Table 1.

The joint examination of the variables "Age," "Vegetable frequency," and "Table number" for both products reveals an overall stability of sensory evaluations across the various studied quality criteria. These results confirm the robustness of sensory evaluations.

After ANOVA analysis, we proceed to examine appreciation score distributions and conduct the Wilcoxon signed-rank test to detect notable differences in specific organoleptic parameters between YALI and TALI products.

Product	Criteria	p-value								
		Odor	Color	Visual texture	Overall appearance	Salt	Bitterness	Acidity	Garlic	Overall taste
YALI	Age	0.592	0.952	0.902	0.688	0.25	0.187	0.00681**	0.695	0.802
	Vegetable frequency	0.347	0.0716	0.358	0.858	0.242	0.0254 *	0.366	0.0689	0.157
	Table number	0.261	0.892	0.364	0.523	0.94	0.667	0.636	0.176	0.0995
	Results	No effect	No effect	No effect	No effect	No effect	No effect	No effect	No effect	No effect

Table 1: Results of ANOVA statistical test analysis performed with R software for YALI product. Different error risks than 5% are defined by the R software during result analysis. The error risk at 1% applies to values marked with a *. Therefore, these results are not significant because the p-value is greater than 0.01. The error risk at 0.1% applies to values marked with **. Therefore, these results are not significant because the p-value is greater than 0.001. **Error risk: 5%, * Error risk: 1%, ** Error risk 0,1%.**

Product	Criteria	p-value								
		Odor	Color	Visual texture	Overall appearance	Salt	Bitterness	Acidity	Garlic	Overall taste
TALI	Age	0,94	0,161	0,0425*	0,109	0,00452**	0,242	0,0286*	0,283	0,278
	Vegetable frequency	0,365	0,177	0,635	0,992	0,359	0,939	0,3	0,705	0,195
	Table number	0,667	0,157	0,0494*	0,0792*	0,9	0,564	0,825	0,332	2,551
	Results	No effect	No effect	No effect	No effect	No effect	No effect	No effect	No effect	No effect

Table 2: Results of the ANOVA statistical test carried out with R software for the TALI product. Different error risks than 5% are defined by the R software during result analysis. The error risk at 1% applies to values marked with a *. Therefore, these results are not significant because the p-value is greater than 0.01. The error risk at 0.1% applies to values marked with **. Therefore, these results are not significant because the p-value is greater than 0.001. **Error risk: 5%, * Error risk: 1%, ** Error risk 0,1%.**

Distribution of appreciation scores and Wilcoxon signed-rank test.

The data in Figure 2 and 3 was obtained using the hedonic test.

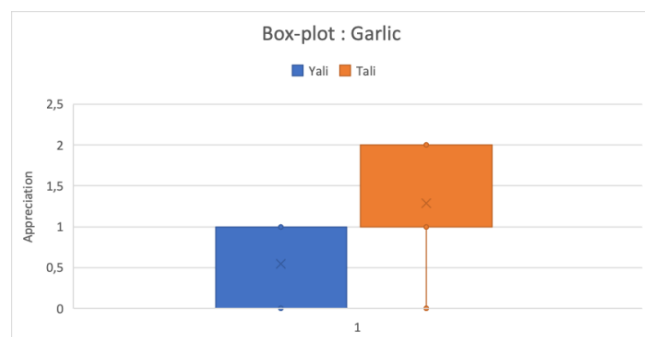


Figure 2: Boxplot comparing the mean distribution of garlic appreciation scores between YALI and TALI products on a sample of 35 people. (0= I don't like at all.1= I don't like.2= I like moderately.3= I like.4= I like well.5= I like a lot)

Figure 2 presents a box plot comparing the distribution of appreciation scores for garlic between the two products (YALI and TALI). It can be observed that the product TALI has a maximum appreciation score of 2 and a minimum of 1. The data for garlic appreciation in the TALI product are asymmetric, with a distribution predominantly located in the upper part of the box plot compared to the mean. On the other hand, the product YALI has a maximum appreciation score of 1 and a minimum of 0.5. Additionally, the data for the YALI product are symmetrically distributed around the mean. This implies that there are an equal number of data points in the upper and lower parts of the box plot concerning the mean. Consequently, it can be stated that the TALI product was more appreciated in terms of garlic compared to the YALI product. Therefore, there is a significant difference in garlic appreciation between the two products.

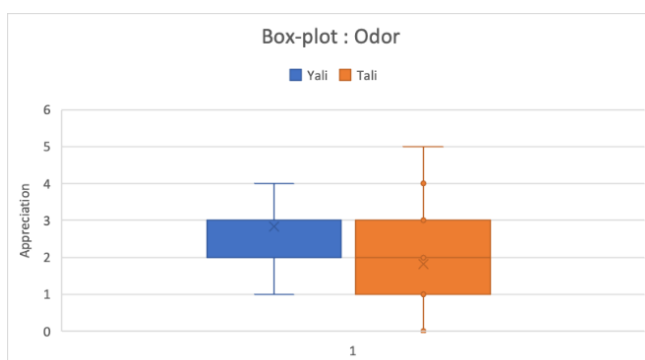


Figure 3: Boxplot comparing the mean distribution of odor appreciation scores between YALI and TALI products on a sample of 35 people.

Figure 3 depicts a box plot comparing the distribution of appreciation scores for smell between the two products (YALI and TALI). It is evident that the TALI product has a maximum appreciation score of 5 and a minimum of 0.

Conversely, the YALI product has a maximum appreciation score of 4 and a minimum of 3. Notably, the box plot for the TALI product is larger than that for YALI, indicating a more heterogeneous distribution. Indeed, the TALI product exhibits a more varied distribution compared to the YALI product. Therefore, it can be concluded that there is a significant difference in the distribution of appreciation scores for smell between the two products.

We focused on comparing our two products based on various organoleptic parameters such as smell, color, visual texture, overall appearance, salt level, garlic content, bitterness, acidity, and overall taste. To achieve this, we conducted a Wilcoxon signed-rank statistical test given the size of our sample, using the R software (Table 3). The chosen significance level is 5%. The p-values for the "smell" and "garlic content" parameters are 0.001 and 1.04×10^{-5} , respectively, both lower than 0.05. This indicates a significant difference between our two products concerning smell and garlic content. Our products are nearly identical in terms of the other parameters: color, visual texture, overall appearance, salt level, bitterness, acidity, and overall taste. This is explained by a p-value greater than 0.05.

CONCLUSION

To conclude, our study outlines a comprehensive protocol for conducting an accessible academic-level hedonic comparison test, emphasizing the importance of careful planning, standardization, and statistical analysis. The flowchart presented in Figure 1 provides a clear roadmap for researchers to follow, ensuring a systematic approach to gather valuable sensory data. The ANOVA statistical tests, as showcased in Tables 1 and 2, delved into potential variations across multiple organoleptic parameters for the YALI and TALI products. The analysis considered variables such as age, vegetable frequency, and table number, revealing overall stability in sensory evaluations. Notably, the absence of significant differences in most criteria underscores the robustness of our hedonic test results. Figure 2, displaying a box plot for garlic appreciation scores, highlights a significant difference between YALI and TALI. TALI outperformed YALI in terms of garlic appreciation, as evidenced by higher mean scores. Similarly, Figure 3, depicting a box plot for smell appreciation scores, reveals significant differences between the two products. TALI exhibits a more varied distribution compared to YALI, indicating diverse preferences among participants.

The Wilcoxon signed-rank test further confirms distinctions between YALI and TALI, with significant differences identified in odor and garlic content appreciation. This supports our earlier findings, emphasizing that certain attributes contribute to varying levels of preference among participants. In essence, our study provides a framework for rigorous hedonic testing, showcases statistical analyses to discern product differences, and unveils specific attributes

that influence consumer preferences. These insights contribute to the field of sensory evaluation, guiding researchers and industry professionals in refining products based on consumer perceptions.

Parameters	p-value	Results
Odor	0,001	Significant difference between the two products
Color	0,29	NA
Visual texture	0,08	NA
Overall appearance	0,08	NA
Salt	0,07	NA
Garlic	1,04⁻⁵	Significant difference between the two products
Bitterness	0,5	NA
Acidity	0,77	NA
Overall taste	0,38	NA

Table 3: Results of the Wilcoxon signed-rank statistical test comparing YALI and TALI using R software. Error risk: 5%, p-value<0.05= significant

DISCUSSION

Our study on hedonic comparison yielded promising results, yet it is crucial to acknowledge inherent limitations. Concerns revolve around managing prerequisites like food allergies and dietary restrictions, where despite efforts to collect information, incomplete responses could impact validity. Furthermore, uneven participant distribution by age poses a potential limitation; balancing age groups could mitigate age-related preference variations. The reproducibility of the hedonic test method is also a central concern, where systematic verification would have strengthened validity. Despite these limitations, our study provides meaningful insights into consumer hedonic preferences. However, identifying these constraints emphasizes the need for a more rigorous methodological approach in future studies.

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The Pleasure of Tasting: A Hedonic Analysis of a Food Product in an Academic Context – Supplementary Material

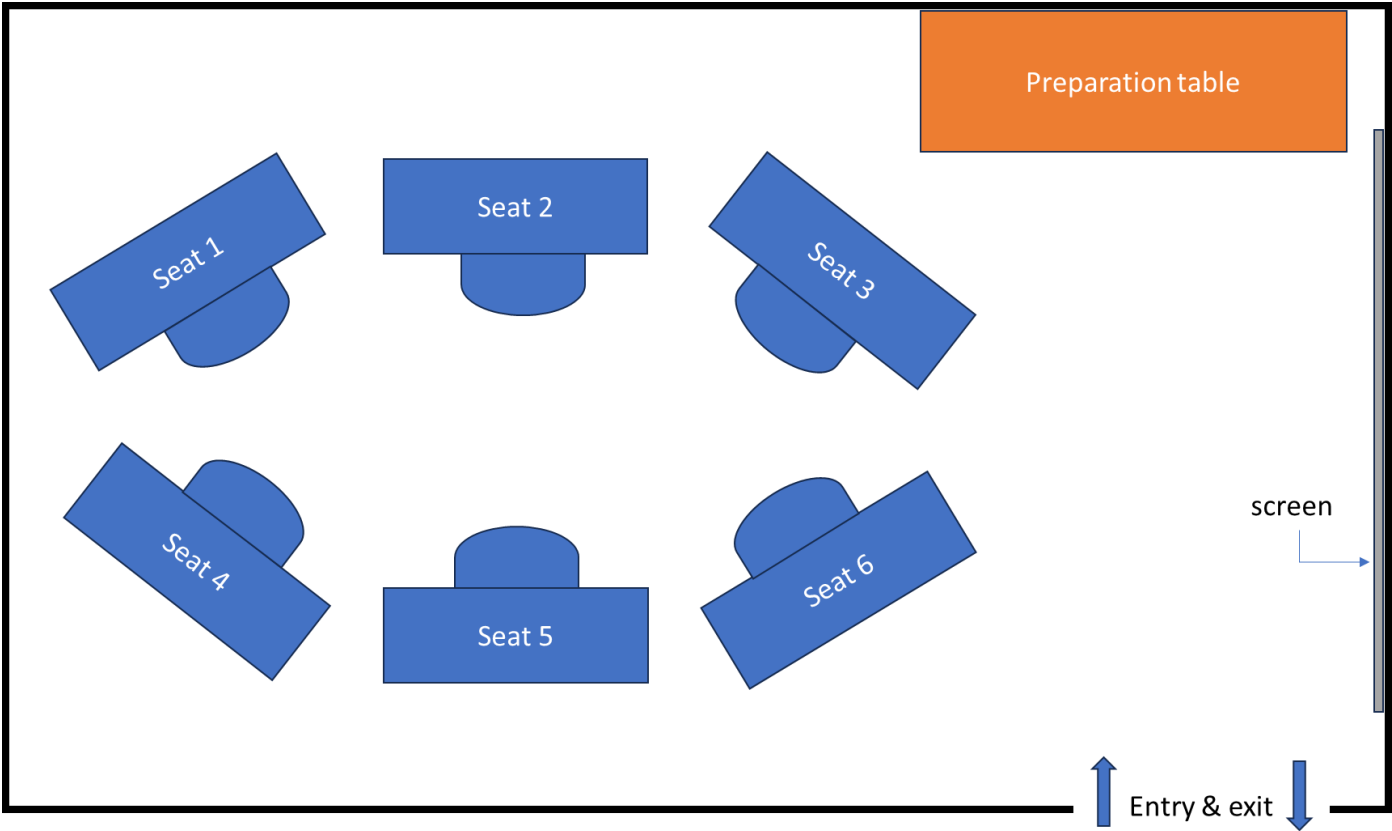
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APPENDIX 1: PLAN OF THE TEST ROOM

APPENDIX 2: FORMS FOR THE TASTING TEST

APPENDIX 1: PLAN OF THE TEST ROOM



APPENDIX 2: FORMS FOR THE TASTING TEST

PLEASE CIRCLE THE ANSWER THAT IS APPROPRIATE FOR YOU.

ONLY ONE ANSWER MAY BE POSSIBLE PER QUESTION.

First of all, let's get to know each other.

The 5D Project student team informed me of the purpose of this questionnaire, as well as how my responses to this questionnaire will be stored and processed. By answering this questionnaire, I authorize the student team to store and process my answers, in an anonymized manner, for research and communication purposes, within the framework of Sup'Biotech's 5D Projects only.

What is your gender?

Man	Woman	Other	I don't want to answer
-----	-------	-------	------------------------

How old are you?

..... Years	
-------------	--

How often do you eat vegetables per week?

Every day	More than 3 times a week	Less than 3 times a week	Never
-----------	--------------------------	--------------------------	-------

Are you a smoker?

Yes	No
-----	----

Have you eaten or drunk anything other than water in the last hour?

Yes	No
-----	----

What table number are you sitting at?

1	2	3	4	5	6
---	---	---	---	---	---

5T7 Product Test

You can now enjoy the 5T7 product presented to you and answer the following questions.

How do you find the smell of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

How do you find the color of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

How do you find the visual texture of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

How do you find the overall look and feel of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

How would you rate the amount of salt in this product?

Trop foot of sel	Enough	Too much salt
------------------	--------	---------------

How would you rate the bitterness in this product?

Amer foot trop	Enough	Too bitter
----------------	--------	------------

How would you rate the acidity in this product?

Too low in acidity	Enough	Too acidic
--------------------	--------	------------

How would you rate the amount of garlic in this product?

Too low in acidity	Enough	Too acidic
--------------------	--------	------------

How do you find the overall taste of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

Does this product have a place for an aperitif?

Yes	No
-----	----

Would you consume this product again?

Yes	No
-----	----

3M8 Product Review

You've finished the questions about the 5T7 product. Don't forget to rinse your mouth. You can now enjoy the second product and answer the following questions.

How do you find the smell of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

How do you find the color of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

How do you find the visual texture of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

How do you find the overall look and feel of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

How would you rate the amount of salt in this product?

Trop foot of sel	Enough	Too much salt
------------------	--------	---------------

How would you rate the bitterness in this product?

Amer foot trop	Enough	Too bitter
----------------	--------	------------

How would you rate the acidity in this product?

Too low in acidity	Enough	Too acidic
--------------------	--------	------------

How would you rate the amount of garlic in this product?

Too low in acidity	Enough	Too acidic
--------------------	--------	------------

How do you find the overall taste of the product?

I don't like it at all	I don't like	I like medium	Like	I like it	I like it a lot
------------------------	--------------	---------------	------	-----------	-----------------

Does this product have a place for an aperitif?

Yes	No
-----	----

Would you consume this product?

Yes	No
-----	----

Cerebro, a Tool for a Better Understanding of Neuroscience

Océane DELVARRE-LEMMER[†], Héloïse DO[†], Quentin DRAGIN[†], Nathan HADDAD[†], Dana KAZMA[†], Lucie LIEU[†], Charlotte ROBERT[†], Timothée TOURNEUX, Yijie LI, Isaline ROY, Julien GRIMAUD*

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Abstract - In France, the teaching of neuroscience in secondary education is still very marginalized and reserved for a few students taking specialized courses in the final year of high school. Moreover, most establishments use outdated software and extremely few material resources exist to recreate conditions that enable hands-on learning of this discipline. All this makes teaching neuroscience a complex task. That's why we decided to create Cerebro, which is a control panel that mimics and shows the action potential in the brain when it is under different kinds of substances. After the development of our product, we obtained a functional product that we were able to test during SupBiotech's science day. Participants were invited to respond to a survey at the end of the activity. They had overall acquired new knowledge and were satisfied with their experience using our product, hence Cerebro could be used as a science awakening and learning tool for a nonscientific public. With these encouraging results we plan to develop hands-on practical associated with our product and thus engage with French educational institutions.

Index Terms - Action potential, Electronic, Neuroscience, Neurotransmitters, Teaching

INTRODUCTION

Active learning is a teaching method that relies on the learner's action and cognitive investment in an activity, leading them to construct their own way of thinking and reflecting on a notion (1). The methods employed can be diverse: reading, writing, speaking, dialogue or reflection. Not all scientists agree on the definition of this method but agree that students "interact with and actually think about (analyze, synthesize, evaluate) the information presented, rather than passively receive it" (2). This method puts the children or students in a position where there will be a active actor learning instead of a passive one. It leans on one of the

principles of the cognitive sciences of the development of the human's brain which is active engagement, curiosity and correcting its errors (3).

Nowadays, teaching neurosciences is done in a theoretical way and is rarely approached in a practical manner. In France, the teaching of neuroscience in secondary education is still very marginalized and reserved for a few students taking specialized courses in the final year of high school. According to the national education program, there is only one chapter about the brain functioning covered in first year of French high school (sophomore year) (4,5). Such teaching is still very rare in non-specialized curricula, as few secondary school teachers are trained in these disciplines. As a result, most students enter higher education without ever having had a chance to get familiarized with this subject. There is currently a lack of tools to study and make a first approach to neurosciences. Most establishments use outdated software to teach these concepts, and extremely few material resources exist to recreate conditions that enable hands-on learning of this discipline.

The study of electrical phenomena inside the brain can help to understand the brain's physiology.

Since the beginning of the twentieth century, no fewer than 14 Nobel Prizes in Science have been awarded for work in neurosciences. In their work [Alan L. Hodgkin](#) and [Andrew Huxley](#), demonstrated that the study of electrophysiology is essential for the progress of medicine and enhancing human health (6). The functioning of the neurons is a complicated concept to approach. Hence it is interesting to develop a model that will be able to reproduce the functioning of the neurons inside the brain, depending on the molecule added. With this in mind, we have created a product that mimics the action potential of the neurons which makes teaching neuroscience easier and allows students to experiment for themselves. Our product is called Cerebro and permits us to teach in an interactive way at school through a ready-to-use

kit. We believe that this field could speak to a greater number of pupils than it currently is. It can also be used as a game for science awakening for people who are not familiar with neuroscience, accessible for everyone, no matter their age. It is known that game-based learning promotes greater enjoyment, which can positively affect deep learning, higherorder thinking, and additional transferable skills (7,8). To illustrate our point, we will present the development of the product in this article, from hardware to software, and results from a survey that has been made during Supbiotech's Science Day where we presented Cerebro as a game. Overall, it could play a major role in raising awareness of the use of neuro-stimulants and narcotics. It would allow the public to visualize the drug's effect on the brain and more particularly on the neurons. This other approach would complete the other awareness campaigns that are already in place.

STATE OF THE ART

A neuron is the smallest living part of the nervous system responsible for transmitting nerve impulses (9). They form an internal communication network that enables us to respond to both internal and external environments to survive (10). Neurons communicate with each other thanks to the synapse which is the area where the signal is carried by chemical messengers (9). The signal generated by the neurons is an action potential, an electric signal that is governed by the all-or-nothing law. In biology, this law refers to whether a response appears following stimulation. If the response appears, the phenomenon recorded always has the same characteristics (11). Some types of molecules can speed the frequency of the signal that goes through the axons of the neurons and some can slow it down. These include neurotransmitters, which are chemical molecules contained in synaptic vesicles that transmit neuronal messages, and neuromodulators, which are chemical molecules that modify or regulate nerve activity by acting on a neurotransmitter (9). Due to the complexity of approaching, visualizing and understanding this subject, we decided to produce a product in the form of an electronic box (Figure 1). The aim of the box is to simulate the observation of neuronal activity at rest and under substances, thanks to scrolling bars on an LCD screen representing action potentials. We have also created a plastic brain to which we have connected fake probes, materialized by wires that are linked to the control box. The control box has buttons which, when pressed, mimic the injection of a substance to excite or inhibit the action potential. The effect of this injection is represented on the control unit's screen, which displays bars that scroll at different frequencies, depending on the substance.

MATERIAL AND METHODS

The production of the casing in its entirety requires a well-oiled mechanism to function. This includes a hardware part, with the assembly of all the elements of the case (3D printed shell, LCD screen, buttons, etc.) and a software part, which involves coding an electronic board to ensure that the various

elements of the case function correctly by giving them clear instructions. This last part is invisible on the finished product, but it is nonetheless essential.

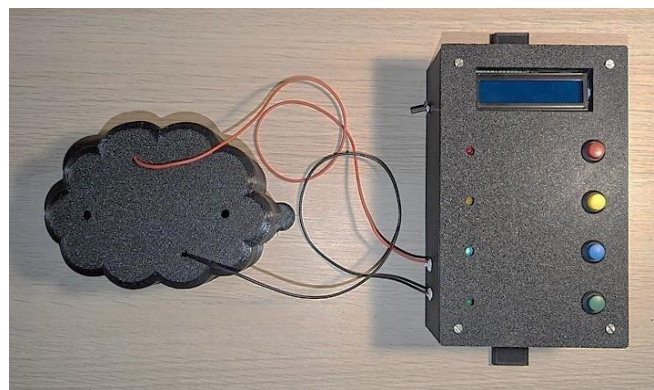


FIGURE 1. PHOTOGRAPHY OF THE FINAL PRODUCT THE 3D PRINTED CONTROL PANEL WITH BUTTONS AND THE BRAIN-SHAPED BOX BOTH CONNECTED TO EACH OTHER WITH CABLES.

I. HARDWARE PART

On the hardware side, we needed several components to build the box. We used common components like cable to drive electricity, LED to make light, buzzer to make noise, push button to allow or not the passage of electricity, resistors to protect the LED and button by reducing the electricity tension. Resistors have different capacities to decrease the electric tension (220Ω and $10k\Omega$). We also needed Printed Circuit Boards (PCBs) to hold all the above-mentioned components, and terminal blocks to connect cables more easily. Another important piece used is an LCD screen that includes 16 pins. A pin is a metallic part which allows the connection to another component. A table with all component part numbers is available in Appendix 1. All components were directly or indirectly linked to an electronic board that operates them. This board is the Arduino® UNO. It is programmable in the Arduino® language, which is a simplification of the C++ programming language. The board had several key components: a microcontroller, an oscillator, several input and output ports, a USB connector, a voltage regulator, an LED indicating the status of the board and a reset button (12).

We had to solder the components to the PCBs to make the whole circuit. To do this, we used a soldering iron and tin. Circuit design on Tinkercad® involves firstly the creation of a theoretical plan, with all the components, and secondly physical assembly.

The circuit is divided into different parts as we can see in the Appendix 2: the LCD screen, the four different signals ways and the brain. Pins of the LCD screen is linked to specific Arduino® pins to receive some information about the pixel to light. The screen is also linked to a potentiometer. It's a variable resistor. With this device, we can manually control the electricity tension to increase or decrease the brightness of the screen. The signal path is a little loop which includes

cables, resistors, push button and led. Each loop is powered by the Arduino® and plugged to a specific pin. When the button is pushed, the Arduino® receives the electricity from a specific loop, and it launches a code linked to the specific pin. The brain is a plate with a resistor, a buzzer and a led. The buzzer and the LED followed the frequency of the emission generated by a lane. This module is outside of the box, it's the brain part.

All the references of the listed components are listed in *Appendix 1*.

II. SOFTWARE PART

On the software side, we loaded the code on an Arduino® board. This code uses an LCD display and buttons to simulate neural action potentials. The buttons are associated with neural scenarios: excitation or inhibition. The bars on the screen represent these potentials, and their movement is controlled by lists of initial positions specific to each scenario. When the control panel is switched on, it displays on the screen a normal action potential, when a button is pressed, the code adjusts the position of the bars according to the scenario selected. The LED lights up when the bars return to their initial position (as soon as the bar appears on the LCD screen), simulating neural activation. The choice of button pressed determines the scenario chosen, and the code updates the positions of the bars on the LCD display in real time according to the scenario in progress. Each button can be associated with a substance depending on the need of the activity. Appendix 3 shows the code we have devised in human language. In our current configuration, signals 1 and 3 refer to an excitatory neuronal scenario, while signals 2 and 4 refer to an inhibitory neuronal scenario.

The code was written with the help of Mr. TOURNEUX Timothée, cyber-security engineer. The code starts by initializing the configurations, such as the pins of the buttons, the LCD screen, and the LED, as well as the starting positions of the bars for each neural model. Then, there is a main loop which continuously monitors the state of the buttons to select the neural model according to the number of buttons pressed. We have also created several functions to meet our specific needs. The *PA_select* function is created and used to display animated bars on the LCD screen which depends on the neural scenario, considering the initial positions specific to each model. This function also turns the LED on or off depending on the position of a bar. We want the user to be obliged to remain pressed on the button, allowing him to maintain his attention, thanks to our conditional function *currentButtonState*. Finally, the *nbButtonsPresses* function is used to count the number of buttons pressed simultaneously, blocking any change of scenario if several buttons are activated. This last function is useful for approximating the conditions of the experiment, i.e one substance injected at a time.

III. PACKAGING

For our product, we decided to print the control panel and the brain-shaped box with a 3D printer from our school SupBiotech. The original plans were made on the software Solidworks by Yijie LI, a CIFRE PhD Student from SPARTHA Medical. Then, we modified them on the online software Tinkercad to create the holes for the screws and buttons. To print the boxes a black ABS (Acrylonitrile Butadiene Styrene, Amazon) filament 1.75mm has been used, which is a highly Resistant Durable 3D Printer Filament. The products were sliced with the software Ultimaker Cura and then printed with the 3D printer (Machine 3D Sidewinder X1 Artillery) from the school SupBiotech (Figure 2).

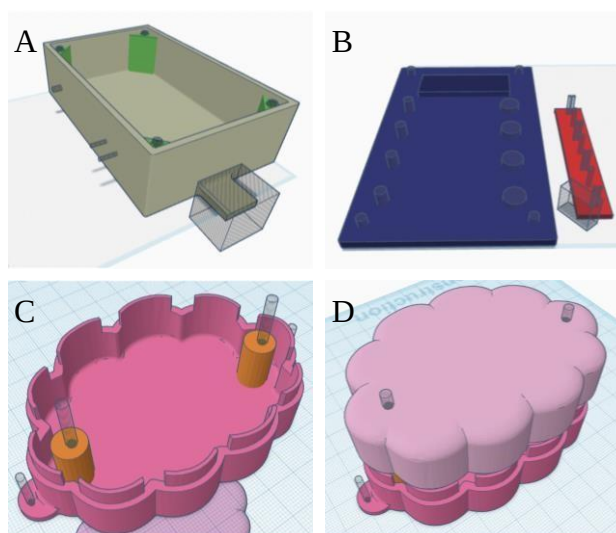


FIGURE 2. 3D PLANS MADE ON TINKERCAD® FOR THE PACKAGING IMAGE A IS THE BOTTOM OF THE CONTROL PANNEL, IMAGE B IS THE LID OF THE CONTROL PANNEL, IN IMAGE C IT'S THE BRAIN BOX BOTTOM AND IN IMAGE D IT'S THE BRAIN LID. THE STRIPES AREA ARE THE HOLLOW ZONE THAT WILL BE THE HOLES OF THE EQUIPMENT.

RESULTS

I. FINAL PRODUCT: CEREbro

Finally, we were able to create a functional device which is a set of 2 boxes (Figure 1): one is the control panel which is composed of 4 buttons that will give a specific message on the LED screen. The other one is a brain-shaped box that needs to be connected to the control panel for the lighting of the whole system. The whole design has been thought to be attractive and to keep the attention focused on it with the colored buttons, the LEDs that light up and by pressing and holding the buttons so that the action potential appears on the screen. We designed the package to be manipulable and safe for children. Concerning the code and the electrical part, they work as we need them to, the buttons are responsive, and the animated bars are displayed. If the cables aren't connected to the brain, then the current doesn't flow, and the box doesn't display anything as required.

II. FIRST TEST OF CEREBRO DURING SCIENCE DAY

Since the product is functional, we used it during the Science Day organized in our establishment on 10/14/2023. The goal was mainly to test our product on the general public that may not have a scientific background and hence determine if it would be suitable for a practical. It aimed to present the Cerebro project through an activity using the final product that teaches basic concepts of neuroscience and how the action potential works. The public was welcomed by explaining shortly the project: explanation of the role of an action potential with a schematic representation of the latter, presentation of a model of the brain with its different zones and observation of onion cells on a captioned photo, to understand what a cell is. After that, the public is invited to visit the box stand. People will find themselves in front of the box, various substance labels (glutamate, adrenaline, adenosine, morphine) and brief definitions of each of these substances that must be placed on the control panel. Those substances were chosen because they are easily explainable by everyday examples (sport, sleep...). For the relevance of the activity, the explications of the impact of the substance on the action potential was different regarding the age of the participant.

People had to observe the box and try to place the substance labels in front of the corresponding button. For them to understand why each substance modified the action potential, they had to follow the instructions given by the team and they were allowed to ask questions. The goal was to press the button, see what happens on the action potential and then link what was observed to the different substances written on the labels.

Finally, once the activity was over, a google form was filled in by the participants to collect data about the workshop. Results of the survey were then assessed and interpreted to determine if our product aimed to teach the participants about neuroscience.

As mentioned previously, because our product is functional, we were able to test it with the public during the Science Day. We collected the results we obtained during the science day. On figure 3A, we determined the level of education of the people who participated in the activity. We can see that most people had a master's degree or more, but people from all kinds of education levels participated in our experiment. On figure 3B, we evaluated the satisfaction level of the activity per person that we asked the participants to grade. Out of 30 people, 23 "loved the activity", and 7 "liked it". None disliked the activity.

On figure 3C, we wanted to assess how our product could improve the level of participants on action potential. Before the activity, the participants had to evaluate their knowledge on the action potential and had to choose between 3 answers to the question "before the science day have you ever heard about the action potential?": "yes and I can explain it", "Yes

but I can't explain it", "Never heard of it". After the workshop, they had to evaluate again their knowledge on the action potential, answering the question "How much do you know about the action potential after the activity?" choosing between 3 levels: "I understood and I can explain it", "I understood but I can't explain it" and "I didn't understand". We then compared the evolution between the first answer given by the 30 participants and their level of knowledge at the end of the experiment. If we analyze the results, we can see that 21 people didn't learn more than they knew at first (blue line and purple line). Those people were the one that already graduated from high school (data not shown). This result is explained by the fact that neuroscience and action potential is taught in high school in France, and they have already heard about it. 4 people knew what the action potential was and couldn't explain what it was at the beginning of the experiment and were able to explain it after the workshop (yellow line). Moreover, 2 never heard of it, understood it during the experiment but were not able to explain it at the end (green line). Finally, 3 people never heard of the action potential before the experiment but understood and were able to explain it after the workshop (red line). Thus, 9 people learned about the action potential and progressed thanks to our activity and 21 people stagnated. All these 21 people already knew what an action potential was before the activity. None regressed.

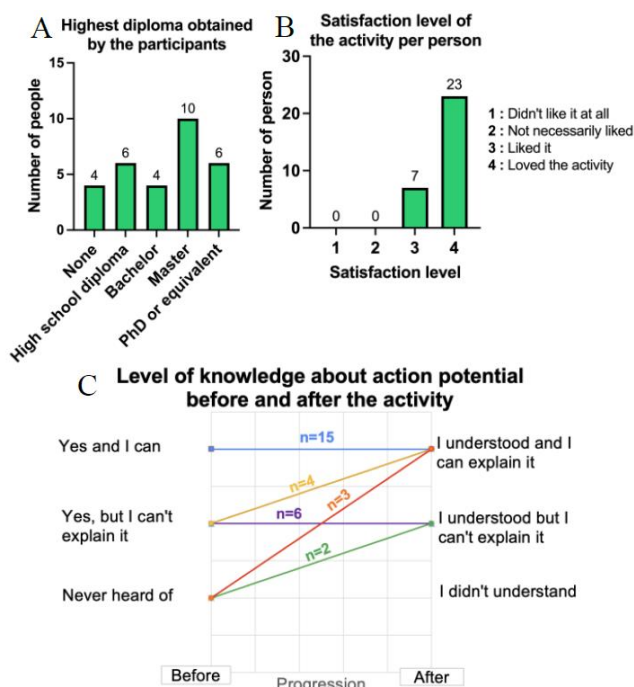


FIGURE 3. RESULTS WE GOT FROM THE PRODUCT TEST WITH HISTOGRAM A SHOWING THE ACADEMIC PROFILE OF THE PARTICIPANTS TO THE CEREBRO WORKSHOP, HISTOGRAM B SHOWING THE SATISFACTION RATE OF THE PARTICIPANTS TO THE CEREBRO WORKSHOP, GRAPH C SHOWING THE EVOLUTION OF THE NEUROSCIENCE LEARNING OF THE PARTICIPANTS FOLLOWING THE CEREBRO WORKSHOP.

Our product seems to have a good impact on the knowledge of the participants about the action potential and be liked by them. Our product would be useful for people who have no or little knowledge about the action potential and to make an initiation in neuroscience. Our goal is to try to test it with the practical sheet we created.

DISCUSSION

Results are interesting, they show that our product is promising because it helps people to increase their knowledge about action potential. However, the public that participated to the experiment was not really the public targeted. Cerebro has been designed to match the educational program of middle school students and most of the public we received were students or teachers and searchers from Supbiotech, which means that they already had some knowledge about neuroscience and action potential. Those people had already heard about the action potential and knew what it was and felt like they didn't progress. Hence, we should conduct more activities or the hands-on practicals with middle school students to have more solid results. That way, we could match the requirements of the French educational system. Due to lack of time, we couldn't do it before publishing the paper, that's why we organized the science day.

Moreover, the packaging of our product still need to be improved. At first, we wanted to put a lightbulb in the brain that will lights up when an action potential is emitted. We wanted to print the brain-shaped box in a transparent cable so that the lightbulb can shine through it but the cable was too difficult to work it, so we took the black cable instead. With more time, we could answer to all the problems mentioned above.

CONCLUSION

In conclusion, our Cerebro device stands as an innovative solution in the domain of education, initiation, and awareness of neuroscience. Through meticulous design and intelligent integration of cutting-edge technologies, our product aims to generate interest and facilitate the understanding of complex concepts related to brain function, such as action potential and the impact of substances on its frequency. Intended for the general public and students from middle school and high school, Cerebro provides a comprehensive, immersive, and educational experience, transforming the learning of this discipline into an accessible and captivating adventure. The positive results received during the Science Day demonstrate the promising impact of Cerebro: participants who had no or little knowledge on neuroscience have overall acquired new knowledge and expressed satisfaction with their experience using our product. These findings highlight its potential to inspire a new generation of cognitive science enthusiasts. We are aware that there is room for improvement and advancement in our device, but we have confidence in its potential among the public. Through this project, we aim to

open doors to a deeper understanding of the human brain. We'd like to stimulate the emergence of future researchers in this field where many remains to be discovered.

ACKNOWLEDGEMENT

We'd like to acknowledge Mr. GRIMAUD Julien who helped us during the whole process of this project, as well as SupBiotech for funding our project and the loan of equipment. We'd also like to thank Mr. TOURNEUX Timothée and Ms. LI Yijie for their help in coding and making the 3D plans respectively. Finally, we'd like to thank Ms. ROY Isaline for her implication in printing our prototypes on the 3D printer.

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Cerebro, a Tool for a Better Understanding of Neuroscience – Supplementary Material

Océane DELVARRE-LEMMER*, Héloïse DO*, Quentin DRAGIN*, Nathan HADDAD*, Dana KAZMA*, Lucie LIEU*, Charlotte ROBERT*, Timothée TOURNEUX, Yijie LI, Isaline ROY, Julien GRIMAUD

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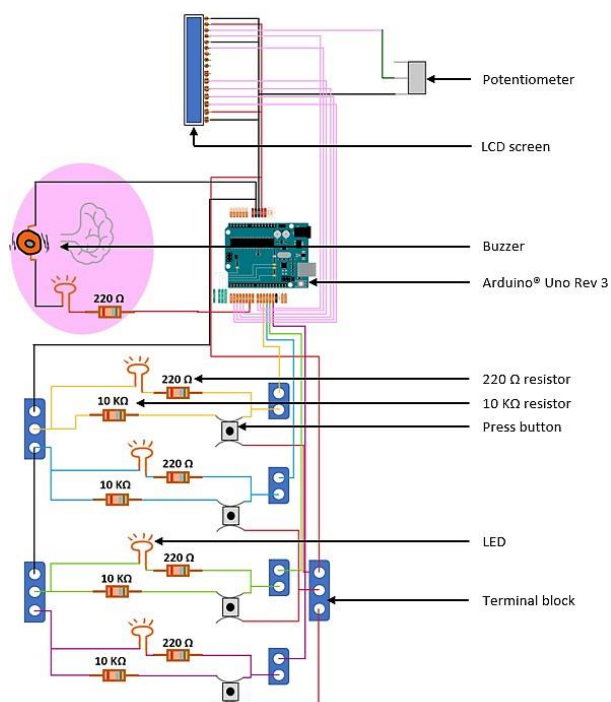
APPENDIX 1: TABLE OF THE COMPONENT USED TO CREATE CEREBRO

APPENDIX 2: SCHEME OF THE ELECTRONICAL CIRCUIT OF CEREBRO. CABLES ARE REPRESENTED WITH LINES IN DIFFERENT COLORS.

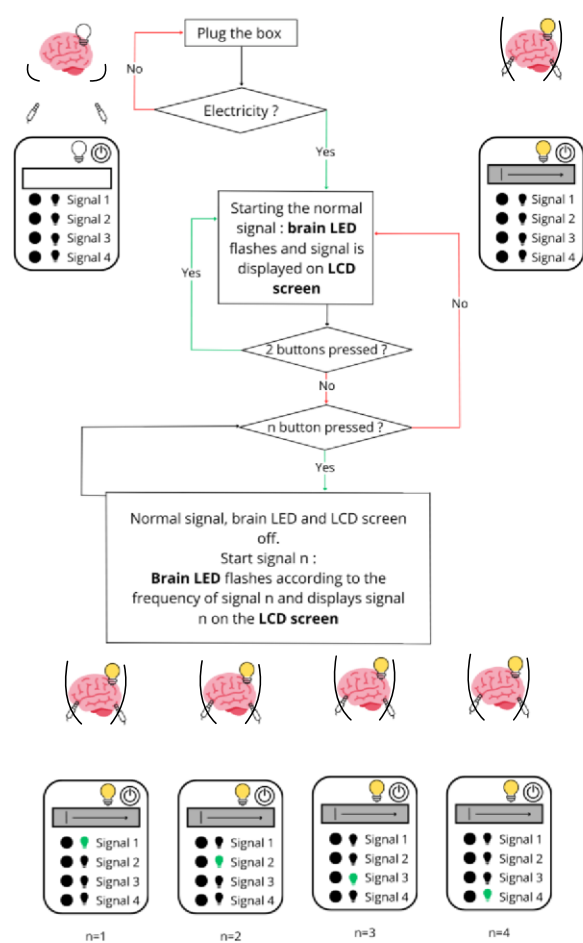
APPENDIX 3: FLOWCHART OF THE CEREBRO FUNCTIONNING

APPENDIX 1: TABLE OF THE COMPONENT USED TO CREATE CEREBRO

Component	Reference product
Arduino® Uno Rev 3	Arduino® Store: A000066
Buzzer	Amazon: B08GQ5FPRP
Cable	Amazon: B07TV4RJHJ
LCD screen	Amazon: B08216LXVQ
LED	Amazon: B07PZ75N67
Potentiometer	Amazon: B086K3GQ5X
Push button	Amazon: B07XRBDXT
Resistor 10 K Ω	Amazon: B016TGBJGM
Resistor 220 Ω	Amazon: B01JJ94UVW
Soldering iron	Leroy Merlin: 67737936
Tin	Leroy Merlin: 63378273
Battery 9V	Amazon: B07P8YV94C
Cable battery	Amazon: B07DK55VMK
PCB	Amazon: B07DK55VMK
Terminal block	Amazon: B0734XYJPM
Pin male and female	Amazon: B08K32Q9Y9

APPENDIX 2: SCHEME OF THE ELECTRICAL CIRCUIT OF CEREBRO. CABLES ARE REPRESENTED WITH LINES IN DIFFERENT COLORS.

APPENDIX 3: FLOWCHART OF CEREBRO FUNCTIONNING



New User-Friendly Computer Program to Teach Spike Sorting

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Abstract – This article addresses challenges in neuroscience education, focusing on spike sorting, a crucial skill in understanding neuronal communication. We introduce "Spiky" an innovative software, and detail its design, features, and integration into a thematic course on olfaction. Utilizing MATLAB® and the technique of Principal Component Analysis (PCA), "Spiky" aims to enhance accessibility and engagement in spike sorting education.

The methodology includes a full exploration of "Spiky" focusing on its user-friendly interface and practical sessions. Despite a small sample size, feedback from participants suggests positive perceptions of clear learning objectives, well-structured content, and balanced workloads. "Spiky" holds promise in transforming neuroscience education, offering a solution to the scarcity of accessible software tools for spike sorting. Acknowledging limitations, including potential bias, subjectivity, and ethical considerations uphold the study's integrity. In conclusion, this study introduces "Spiky" as a practical innovation, signaling a shift towards more effective and engaging neuroscientific education. The positive feedback and thematic course approach indicate potential for broader implications in addressing challenges in electrophysiology education.

Index Terms – Biology, Electrophysiology, Neuroscience Education, PCA, Spike Sorting, Software Development

INTRODUCTION

In the intricate tapestry of the human brain, neurons stand as the fundamental building blocks of our cognitive universe. Composed of a complex web of dendrites, axons, and a cell body, neurons are the messengers of our thoughts, transmitting information through the brain's vast neural networks (Les cellules nerveuses, n.d.). What makes neurons even more remarkable is their unique ability to transmit electrical signals, also called nerve impulses. It is a property

that has fascinated scientists and researchers for centuries (Moldovan et al., 2022; Nagwa, n.d.).

At the core of this phenomenon lies the concept of the action potential, which is the essence of neuronal communication. An action potential is a brief but crucial event in a neuron's life, characterized by the change of polarity going through the membrane (Grider et al., 2023). Understanding the complexity of action potentials, from their initial depolarization to the subsequent repolarization, is pivotal in unraveling the mysteries of the brain and its complex functions (Grider et al., 2023).

Electrophysiology, the science of studying electrical activity within living organisms, provides the key to decoding the enigma of neuronal signals (Duprat, n.d.; Obien et al., 2015). It involves the use of electrodes to record the electrical activity of neurons, opening a window into the brain's inner workings (Hai et al., 2010; Soscia et al., 2020). The analysis of these signals is a profound endeavor, leading us to the realm of spike sorting, a crucial process in extracting meaningful information from the cacophony of neuronal data (Quiroga, 2007).

Among all of the possibilities to analyze a signal, spike sorting, is the way of distinguishing individual neurons' electrical signals from a complex mix of recordings. This meticulous process requires sorting and categorizing these spikes based on their unique characteristics, such as amplitude, duration, and shape. By doing so, researchers can cleanse the signals to understand how different neurons contribute to brain function.

While the field of electrophysiology holds immense promise, the lack of accessible software for teaching spike sorting remains a significant challenge (AlphaOmega, n.d.; Nick et al., 2013). Today, people who would like to study neuroscience often find themselves struggling with limited resources when it comes to mastering this complicated skill

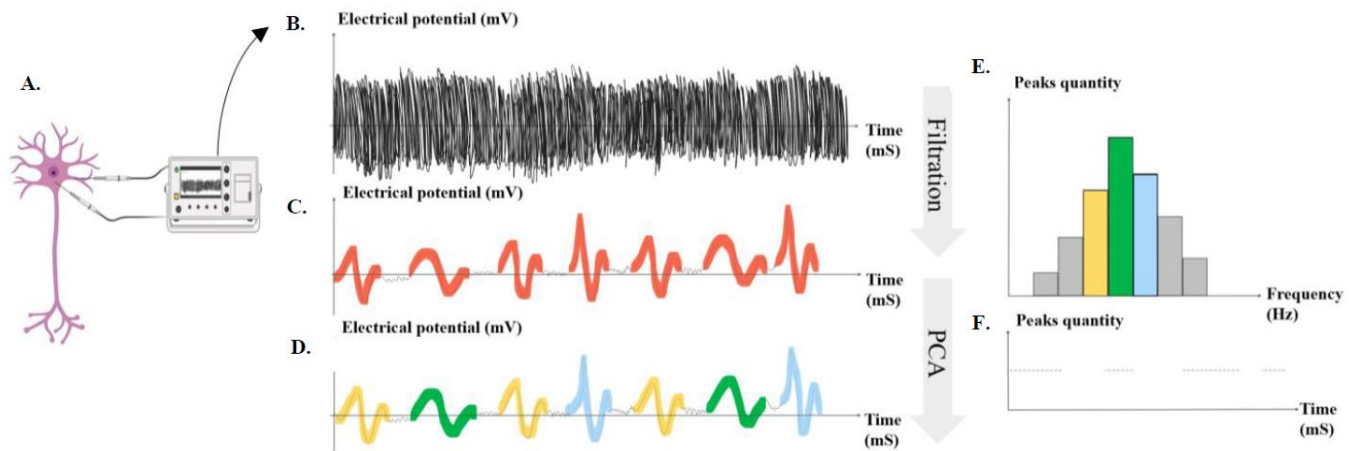


FIGURE 1: GENERAL OUTLINE OF EXTRACELLULAR SIGNALS RECORDING IN NEURONS AND THEIR ANALYSES. A. RECORDING EXTRACELLULAR ELECTRICAL SIGNALS FROM NEURONS. SIGNALS ARE AMPLIFIED AND BAND-PASS FILTERED. B. RAW SIGNAL FROM THE NEURONAL RECORDING. C. SIGNAL AFTER FILTRATION AND EXTRACTION OF COLORED PEAKS. D. DIFFERENTIATION OF PEAKS CORRESPONDING TO THREE DIFFERENT NEURONS (YELLOW, GREEN, AND BLUE). E. QUANTITY OF DIFFERENT TYPES OF PEAKS. F. TIME DISTRIBUTION OF THE PEAKS.

(High-Density MEA Technology - MaxWell Biosystems, n.d.; Qian et al., 2019). Current methods of teaching electrophysiology lack the comprehensive tools necessary for students to grasp the complexities of spike sorting and the existing one are not user-friendly (ESPCI, n.d.; Metrodis, n.d.; Travaux Pratiques - Analyse En Composantes Principales — Cnam – UE RCP208, n.d.). Most of them are not accessible, or they are costly. Altogether, it limits the access to researchers' community while there are at least 20 masters in France who need easy tools to teach spike sorting. (Master mention neurosciences, n.d.)

Enter "Spiky", an innovative software designed to bridge the gap in spike sorting education. As we stand on the precipice of a new era in neuroscience, "Spiky" symbolizes a valuable resource for both students and educators. This innovative tool not only simplifies the process of analyzing neuronal signals but also empowers instructors to teach spike sorting with precision and efficiency, paving the way for a deeper understanding of the brain's electric symphony. This tool is coupled with a course on spike sorting and a tutorial that promise to revolutionize the way neurons and their action potentials are studied.

In this article, we will explore the challenges faced in teaching this critical discipline today and introduce you to the groundbreaking "Spiky" software, poised to revolutionize the way we learn and understand the complex art of spike sorting.

LITERATURE REVIEW: NEURON THEORY, SPIKE SORTIN

The neuron is the basic unit of the nervous system, responsible for transmitting information through the neural network. It consists of a cell body, an axon, and dendrites. When the neuron is activated, it generates an electrical signal called an action potential also named spike (Kandel & Squire, 2000). It is a brief fluctuation in the electrical potential across

the neuron's cell membrane, triggered by electrochemical signals from synapses (BEAR et al., 2020).

Electrophysiology, a branch of biology, employs techniques such as electroencephalography (EEG) and intracellular electroencephalography (patch-clamp) to record and study these electrical signals (Gardner-Thorpe, 2000). These techniques provide valuable electrophysiological data to understand how neurons communicate with each other.

However, the massive amount of data generated by these techniques requires sophisticated tools for analysis. This is where spike sorting comes in, a process aimed at identifying and classifying the different recorded spikes (Quiroga et al., 2004). Multiple neurons can contribute simultaneously to the recorded signals, and spike sorting helps unravel these individual electrical activities in an overall recording.

Spike sorting techniques involve the use of advanced algorithms that analyze spike waveforms, amplitudes, and other features to group them accurately (Harris et al., 2000). The goal is to reconstruct the individual activity of each neuron in a recording, which is essential for understanding complex neural networks.

The first step of spike sorting, as shown in figure 1a., involves detecting spikes or action potentials in the electrophysiological recording. This is typically done by setting a threshold above the background noise, thereby identifying moments when neuronal activity surpasses a certain amplitude threshold. It can be defined manually or automatically depending on the amount of data to be processed. It is based on the standard deviation of the signal (Pedreira et al., 2012). To better visualize the steps of spike sorting, the general outline of extracellular signal recording in neurons is shown in figure 1. In this figure, the signals recorded in figure 1a are amplified (fig 1 b.), filtered (fig 1 c.),

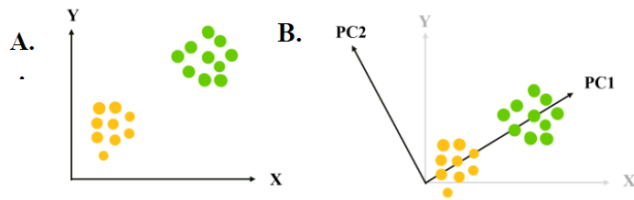


FIGURE 2: PRINCIPAL COMPONENT ANALYSIS PRINCIPLE. A. CLUSTERS IN THE ORIGINAL GRAPHIC. POINTS FROM THE 2 CLUSTERS (GREEN AND YELLOW) ARE HARD TO DIFFERENTIATE. B. CLUSTERS AFTER PCA, EASY TO DIFFERENTIATE THANKS TO THE COORDINATE CHANGES.

and clustered (fig 1 d.). The peaks highlighted in yellow, blue, and green correspond to individual neurons. They are distributed over frequency and time (fig 1e,f.).

Manual sorting of action potentials is laborious and timeconsuming. Therefore, methods based on calculations and algorithms, such as PCA or the "wavelet" technique, are preferred for their efficiency (Hulata et al., 2002; Letelier & Weber, 2000; Quiroga et al., 2004).

PCA, widely used in various fields including biostatistics, provides a method for projecting multidimensional data into a lower-dimensional space while retaining the maximum amount of initial information (Wold et al., 1987). This dimensionality reduction facilitates data processing and interpretation like shown in figure 2. In this figure, data are separate in two different clusters that can be more easily

distinguished after PCA. We can then compare the data based solely on their coordinates on the abscissa.

PCA allows exploring correlations between variables, obtaining uncorrelated factors for further analysis, and visualizing observations in a two or three-dimensional space. These advantages are particularly valuable for spike sorting. The figure 3 illustrates a sequence of steps in the analysis of neuronal signals. It begins with the acquisition of these signals in electrical potential over time. These data are then organized into a matrix of electrical potentials over time and signals (fig 3 b.). They are used to generate a multi-dimensional graph, representing the signals over time (fig 3 c.). This graph, when modified by PCA, offers an alternative visualization where the most significant variations are emphasized. Focusing on these principal components (PC1 and PC2), another graph is produced, illustrating PC2 against PC1 (fig 3 d.). Finally, this representation is used to select and separate data clusters, displaying their electrical potentials over time (fig 3 e,f.). Each step in this analysis contributes to a deeper understanding of the structure, trends, and relationships between neuronal signals, providing valuable insights for neuroscience research.

After the PCA, the crucial step of clustering begins. This step aims to form homogeneous groups of action potentials, presumably originating from the same neuron. This task can be accomplished through methods such as Hierarchical Agglomerative Clustering (HAC) or the k-means method,

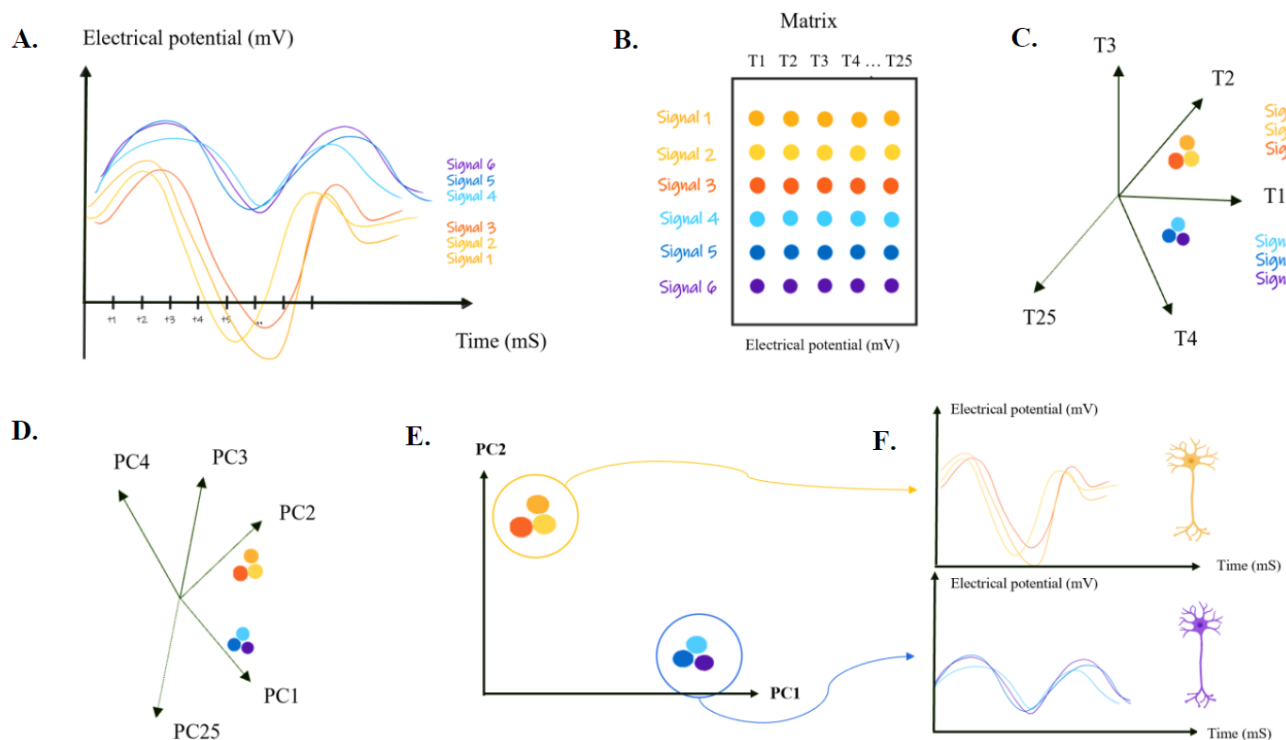


FIGURE 3: PRINCIPAL COMPONENT ANALYSIS IN SPIKE SORTING. A. ACQUISITION OF ELECTRICAL POTENTIAL SIGNALS OVER TIME. B. MATRIX OF ELECTRICAL POTENTIALS OVER TIME AND SIGNALS. C. MULTI-DIMENSIONAL GRAPH OF SIGNALS OVER TIME. D. MULTI-DIMENSIONAL GRAPH OF SIGNALS OVER TIME MODIFIED BY PCA. E. GRAPH OF PC2 AGAINST PC1. F. SELECTION AND SEPARATION OF CLUSTERS AND DISPLAY OF THEIR ELECTRICAL POTENTIAL OVER TIME.

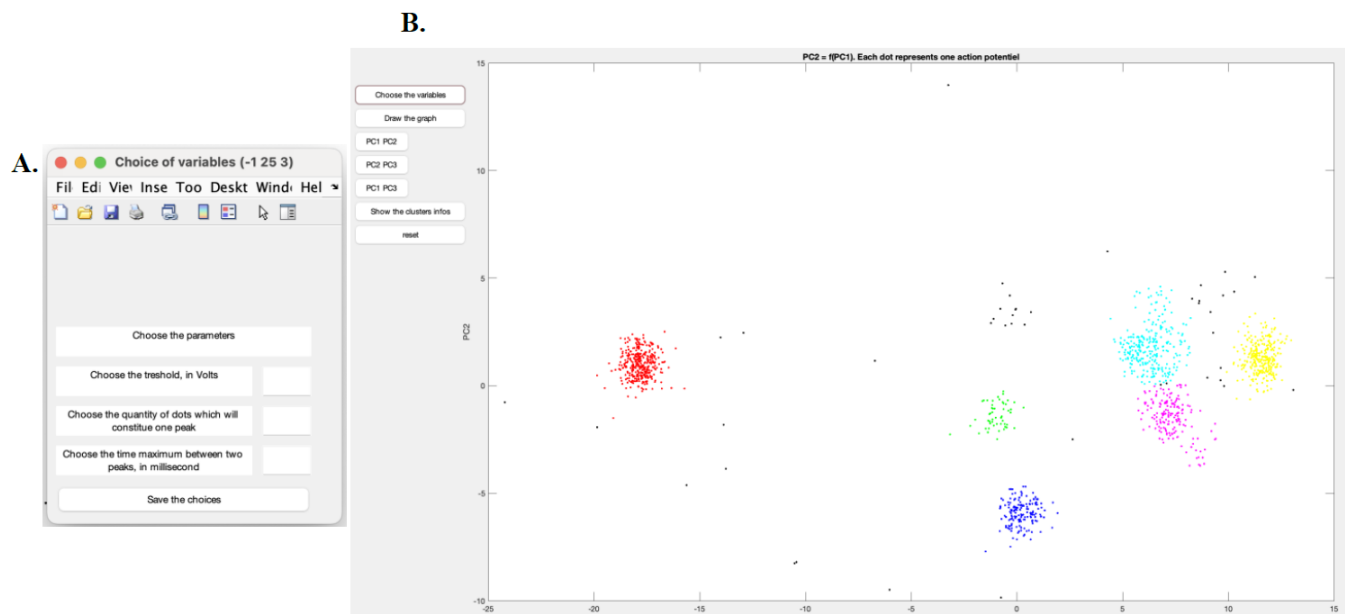


FIGURE 4: INTERFACE OF “SPIKY”. A. POP-UP WINDOW ALLOWING TO INDICATE SPECIFIC ANALYSIS VARIABLES SUCH AS THRESHOLD, PEAK WIDTH, AND REFRACTORY PERIOD. B. WINDOW DISPLAYING THE DIFFERENT SOFTWARE BUTTONS AND SHOWING THE SPIKE CLUSTERS ACCORDING TO THE PCA AXES. EACH COLOR REPRESENTS A CLUSTER.

although the manual delineation of clusters is most often used, despite being recognized as time-consuming and subject to some human bias (Quiroga et al., 2004). Each cluster formed is hypothetically associated with the specific electrical activity of a given neuron, marking a crucial step in the spike sorting process. These clustering methods efficiently organize the data and associate them with distinct neuronal entities, facilitating the analysis of neuronal activity within complex networks (Harris et al., 2000)

METHODOLOGY PART I: SOFTWARE DESIGN, PRINCIPLE, AND USAGE

I. SOFTWARE DESIGN AND IMPLEMENTATION

To fulfill the specific requirements of our software, we selected MATLAB R2022a, version 9.12.0.1184302, renowned for its robustness in signal processing and numerical analysis (Weeks, 2010). The core of our approach is user interaction, aiming to enable intuitive manipulation for students. The interface offers a visual representation of the signal with user-friendly features like customized axes, zoom, and simplified navigation.

A dedicated program was developed to distinguish action potentials from artifacts by identifying background noise and extracting local minima. This approach enhances signal navigation, providing an interactive and educational experience.

Since the signal emanates from different neurons, our software integrates the PCA method, for precise identification (Principal Component Analysis of Raw Data - MATLAB®

PCA - MathWorks France, n.d.). This accommodates the need for an accessible course for undergraduate students without prerequisites. After peak identification, algorithms were implemented to extract key features of action potentials, meeting the illustration needs of a neurology course, including height, mean, instantaneous, and mean frequency.

II. USAGE OF THE SOFTWARE

The software offers an interactive and educational experience for understanding spike sorting, combining analytical power with user-friendliness. Users access the interface via MATLAB®, where a pop-up window, showed on figure 4 a., guides essential information entry for analysis (threshold, number of values, maximum time). Raw data (data) and acquisition frequency (fs) are imported via a dedicated document.

Upon clicking "Compute the data" on figure 4 b., the software performs preliminary calculations, followed by graphical representation of raw data with "Draw the graph". Buttons such as "PC1 to PC2," "PC2 to PC3," and "PC1 to PC3" enable exploration of point clouds based on different principal component combinations. "Select points" allows manual point cloud selection via a polygon.

"Draw graph for the neurons" opens a window per cluster, showed in figure 5. It provides a crucial visual insight into the selected cluster on the software interface, enhancing the understanding of neuronal activity by displaying graphically selected peaks (fig 5 a.), interspike interval distribution (fig 5 b.), and temporal distribution of associated peaks (fig 5 c.).

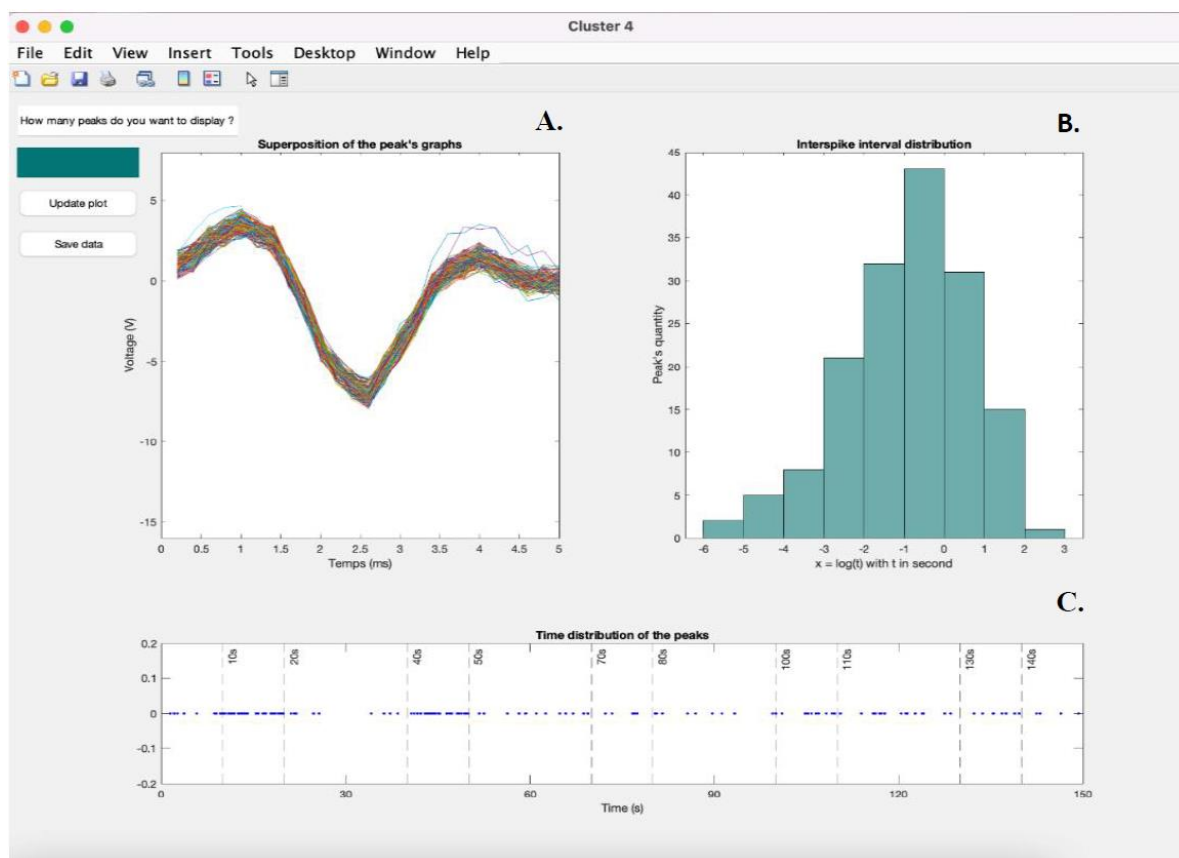


FIGURE 5: INFORMATION ABOUT THE SELECTED CLUSTER DISPLAYED ON THE USER INTERFACE OF "SPIKY" - MATLAB. A. SUPERPOSITION OF THE PEAK'S GRAPHS: EACH DOT REPRESENTS THE VOLTAGE IN VOLT AS A FUNCTION OF THE TIME IN SECOND. THE DOTS ARE CONNECTED BY A LINE IN A WAY AS EACH LINE REPRESENTS A SPIKE. ALL OF THE PEAKS OF THE SELECTED CLUSTER ARE OVERLAID ON THE SAME GRAPH. B. INTERSPIKE INTERVAL DISTRIBUTION: LOGARITHMIC GRAPHIC OF THE INTERSPIKE INTERVAL WHICH IS THE TIME BETWEEN EACH PEAK OF THE SELECTED CLUSTER. EVERY BAR REPRESENTS THE PEAK QUANTITY AS A FUNCTION OF THE LOGARITHM OF THE TIME IN SECOND. C. TIME DISTRIBUTION OF THE PEAKS OF A CLUSTER: EACH DOT REPRESENTS A PEAK ON THE Y-AXIS (TIME IN SECONDS), ILLUSTRATING THE TEMPORAL DISTRIBUTION OF PEAKS CAUGHT BY THE ELECTRODE. STIPPLES INDICATE SPECIFIC TIME INTERVALS CORRESPONDING TO EXTERNAL STIMULATIONS ON THE SUBJECT (AT 10, 20, 40, 50, 70, 80, 100, 110, 130, AND 140)

"Spiky" offers the ability to save crucial information by clicking on the "Save Data" button. This functionality allows users to record important details generated during the analysis. When activated, the "Save Data" option ensures that the analytical results are stored for future reference or analysis. This feature enhances the software's utility, providing users with a convenient way to preserve their findings and facilitate seamless data management within the educational context.

METHODOLOGY PART II: COURSE / LAB COURSE

I. LAB COURSE

In the pursuit of advancing educational tools for higher education, our study focuses on the development and evaluation of a specialized piece of software designed for both students and professors in this academic setting. The aim is to address existing gaps in the educational landscape, particularly in the field of neuroscience. Our approach involves a targeted selection of participants and a carefully crafted curriculum centered around the intriguing theme of olfaction.

Our study directly engaged students aged 20-25 in higher education and educators from scientific institutions. The selection criteria were based on the potential interest of these individuals in utilizing our software to explore neuroscience concepts in an interactive manner. This deliberate approach aimed to ensure a diverse yet focused participant pool.

The primary objective of our study was to showcase and assess the efficacy of our educational software. Recognizing the need for a thematic focus, we chose to delve into the fascinating world of olfaction. Through meticulous research, we constructed a comprehensive curriculum, covering fundamental neurology concepts, neural signal reception using the olfactory bulb. Participants, equipped with neuronal recording devices, developed software, and detailed practical session sheets, engaged in a task to identify the specific odor stimulating their olfactory organs. They were given tables associating the molecules and the smells such as rose, lemon, etc. The figure 6, is one of the figures presented in our practical session sheet. It shows a general outline of the delivery of molecules from nose to brain. Molecules are first inhaled (fig 6 a.), and then are collected through a certain path

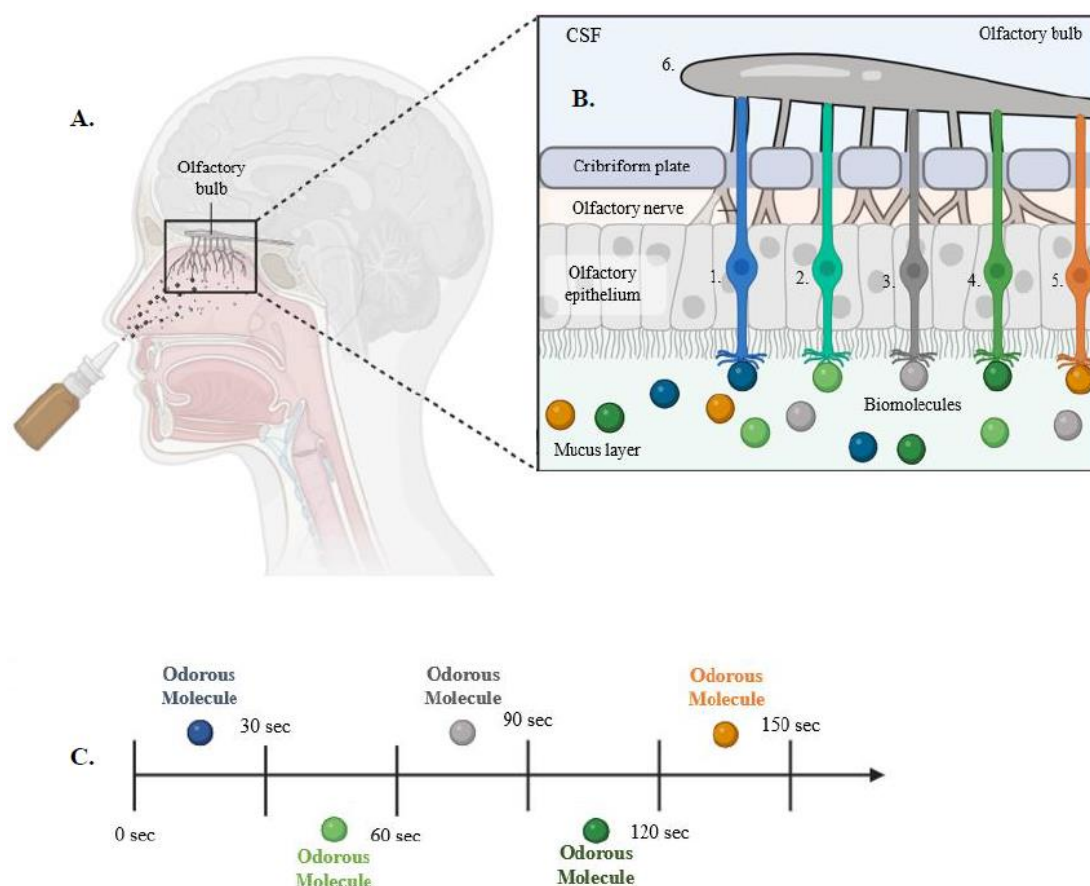


FIGURE 6: NOSE TO BRAIN BIOMOLECULE DELIVERY. A. HUMAN OLFACTORY SYSTEM EXCITED BY ODORANT MOLECULES THAT REACH THE OLFACTORY RECEPTORS IN THE NASAL CAVITY AND TRIGGER NERVE SIGNALS. B. THE SIGNALS ARE THEN TRANSMITTED TO THE OLFACTORY BULB IN THE BRAIN, WHICH INTERPRETS THEM TO PERCEIVE SMELLS (CSF: CEREBROSPINAL FLUID). AN OLFACTORY BULB (6.) IS INVOLVED IN ASSOCIATING SMELLS WITH MEMORIES AND EMOTIONS, CONTRIBUTING TO OUR PERCEPTION OF AND RESPONSE TO DIFFERENT ODORS. THE CRIBRIFORM PLATE ALLOWS THE TRANSMISSION OF OLFACTORY INFORMATION CAPTURED IN THE NASAL CAVITY TO THE BRAIN, WHERE IT IS THEN INTERPRETED TO PERCEIVE AND IDENTIFY ODORS. THE OLFACTORY NERVES, (1. 2. 3. 4. 5.); NEURONS EXCITED DURING THE EXPERIMENT. THE OLFACTORY EPITHELIUM IS THE STARTING POINT OF THE ODOR PERCEPTION PROCESS IN THE OLFACTORY SYSTEM. C. EXPECTING RESULTS OF THE PRACTICAL WORK. TEMPORAL DISTRIBUTION OF ODORANT MOLECULES. EACH COLOR ON THE FRIEZE REPRESENTS THE MOLECULE IDENTIFIED AS A FUNCTION OF TIME.

(fig 6 b.). Finally, figure 6 c., shows in which order the molecules were smelled for our practical work.

To further gauge comprehension, a Kahoot-based quiz was administered at the end of the session, serving both as a reinforcement tool and a data collection method.

To assess the impact of our course, we conducted test sessions with both students and professors. A detailed questionnaire was employed to gather insights into the participants' perspectives. The questionnaire featured two main sections: an evaluation of the pedagogical content and, optionally, an assessment of the presenters. The latter was included to discern whether any limitations in the learning experience were attributable to the presenters or the presentation format. Our report focuses on the results related to the content evaluation (fig 7.), with additional insights into the presenters' assessment available in the supplementary data.

DATA ANALYSIS

In the pursuit of scientific rigor, our study acknowledges the limitation posed by a relatively small sample size, precluding the execution of robust statistical analyses. This inherent constraint necessitated the omission of formal statistical tests in the analysis phase. All the same, 6 people took part in our test, which is still an acceptable number to draw any real conclusions from, as we're talking about the opinions of different students and therefore different needs which, on a class scale, can bring benefits to everyone (Hackshaw, 2008).

In interpreting the feedback garnered from participants, observed in figure 7, several key themes emerge, shedding light on the efficacy of our educational approach. Firstly, the clarity of defined objectives resonates as a significant factor. Students expressed a distinct understanding of what was expected of them by the course's conclusion, signaling the success of well-defined learning outcomes.

Furthermore, the well-prepared and structured content emerged as a pivotal element contributing to the overall positive experience. The ease with which participants could follow and comprehend the course material underscores the importance of meticulous preparation and logical structuring of educational content.

A crucial aspect contributing to the positive reception of our course was the adaptability of the workload. Participants conveyed a sense of appreciation for a workload that was perceived as appropriately balanced, avoiding overwhelming density. This sentiment not only fosters a conducive learning environment but also indicates the facilitation of active student engagement.

Notably, the participants reported a genuine sense of involvement in the learning process. A successful course, according to their perspective, is one where sustained attention is maintained, and our sessions evidently achieved this criterion. The participants' perceived level of engagement serves as a testament to the effectiveness of our instructional design in capturing and retaining student interest.

DISCUSSION AND LIMITATION OF THE STUDY

To uphold the ethical standards of our study, participants' informed consent was sought. This encompassed their agreement not only to partake in the educational session but also to evaluate its efficacy. Furthermore, we took measures to anonymize the results to foster an environment where students and professors felt at liberty to provide candid feedback. Confidentiality was rigorously maintained until the publication of this paper, with no traceability to the individuals who contributed information.

Despite the inherent limitations, we anticipate predominantly positive outcomes from our analysis. As students ourselves, we believe in our ability to discern the needs of fellow students. However, we acknowledge a potential challenge in evaluating the comprehensiveness of the content from a professor's perspective. Uncertainty looms over whether a more detailed guide is requisite for educators, an aspect that might emerge as a highlighted deficiency. We remain open to improvements, recognizing the difficulty of maintaining objective critique when intimately familiar with the subject matter.

It is crucial to note the potential bias present in our results, as a significant proportion of participating students were acquaintances. While efforts were made to uphold objectivity, this inherent association raises the possibility of unintentional bias. Despite their objectivity, friends evaluating the educational content may inadvertently introduce a subjective element, affecting the overall relevance and impartiality of our results.

In navigating the intricate landscape of data analysis within the confines of a limited sample, our study conscientiously

considers the ethical dimensions of participant involvement. While acknowledging potential biases, we strive to glean valuable insights into the effectiveness of our educational tool. The forthcoming results, while tempered by the study's constraints, aim to contribute meaningfully to the ongoing discourse on innovative approaches to neuroscience education.

CONCLUSION

In summary, this article has elucidated the challenges in traditional neuroscience education, emphasizing the intricate nature of neuronal communication and the indispensable role of spike sorting. The study has introduced “Spiky” as an innovative solution, addressing the limitations associated with conventional teaching methods.

Our methodology involved a comprehensive exploration of the software's design, features, and integration into a specialized course centered around olfaction. The incorporation of PCA in “Spiky” and its user-friendly interface were key focal points in our approach. Additionally, the evaluation of the software's impact on students and educators provided valuable insights, despite the acknowledged limitations of a small sample size. The significance of this study lies in the potential of “Spiky” to reshape the landscape of neuroscience education. By simplifying the process of spike sorting and making it more accessible, “Spiky” contributes to fostering inclusivity and engagement in the study of neuronal communication. The thematic approach of the course, combined with practical sessions, represents a departure from traditional methods and offers a dynamic learning experience.

Moreover, the initial success of “Spiky” opens several avenues for future research and development. Longitudinal studies on the learning outcomes of “Spiky” could provide deeper insights into its long-term educational value (Bloom, 1968). Expanding its scope of application to various neurological areas could enhance the versatility and comprehensiveness of this educational tool (Amaral et al., 2007). Collaborations with neuroscientific institutions could also enrich students' learning experience with practical insights into current research (Feldon et al., 2017). Continuous improvements based on user feedback will ensure that “Spiky” remains relevant and effective in the evolving field of neuroscience education (Hattie & Timperley, 2007). The implications of our study extend beyond the immediate context, signaling a shift towards more effective and accessible neuroscience education. “Spiky” holds promise in equipping aspiring neuroscientists with the necessary skills to decipher the complexities of the brain's electric symphony. This is particularly relevant in an era where advancements in technology and an expanding understanding of neuroscience call for innovative educational tools.

In conclusion, our study underscores the importance of adapting teaching methods to meet the evolving demands of

neuroscience education. “Spiky” stands as a practical and impactful innovation, offering a solution to existing challenges in spike sorting education. Its potential to enhance accessibility and engagement signifies a step towards democratizing knowledge in the field, with implications for both students and educators striving to navigate the intricate world of neuronal communication.

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ETHICS STATEMENT

The authors declare that they have no conflict of interest. This article does not contain any studies involving animals or human participants performed by any of the authors.

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SUPPLEMENTARY DATA

Supplementary data to this article can be found online at: [link here](#).

New User-Friendly Computer Program to Teach Spike Sorting – Supplementary Material

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FIGURE S1: GENERAL OPINION OF THE STUDENT ABOUT THE TEACHERS/DEMONSTRATORS

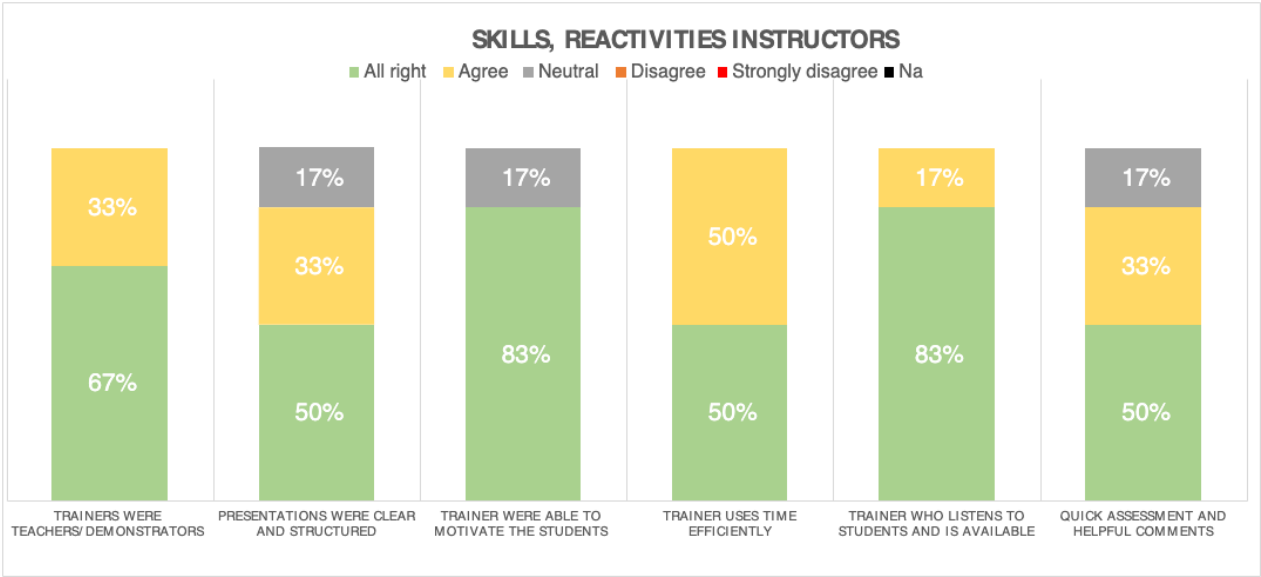


Figure S1 : General opinion of the student about the teachers/demonstrators

The questions were: “Trainers were teachers/demonstrators”, “Presentations were clear and structured”, “Trainer were able to motivate the students”, “Trainer uses time efficiently”, “Trainer who listens to students and is available”. The person had to assign a mark between 1 to 5 to each question. 1 went for “strongly disagree”, 2 for “disagree”, 3 for “neutral”, 4 for “agree” and 5 for “completely agree”.

A Cost-Effective Approach to Bioreactor: Automation With pH Monitoring

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Abstract - The AutoBioX project introduces an innovative bioreactor featuring integrated pH measurement through an Arduino microcontroller. Scientifically, it ensures meticulous control of environmental parameters crucial for cell culture and biomolecule production. The modular design accommodates volumes of 2 to 5 liters, strategically positioned for early-stage experimentation and optimization. The SWOT analysis underscores its strategic market positioning, particularly in the realms of bioprocessing and biotechnological research. Future avenues involve potential expansions for multi-parameter measurements and enhanced automation. In summary, AutoBioX marks a revolutionary advancement in bioprocessing technology by seamlessly integrating scientific precision with market-driven practicality. This fusion promises transformative effects in both biotechnological research and industrial applications.

Index Terms - Arduino, Automation, Bioreactors, Coding, Cost-effective, pH-monitoring.

INTRODUCTION

Bioreactors play a central role in the field of bio-industries due to their ability to provide a controlled environment suitable for the growth of microorganisms, plant and animal cells, or the production of biological compounds.[1] Indeed, they allow meticulous control of environmental conditions that directly influence the cellular growth of various organisms. For example, industries can control pH, temperature, oxygen levels, and so on, there are also 'open-style' or less controlled bioreactors that serve distinct purposes. This meticulous control enables industries to maximize their production of specific biomolecules of interest. It also helps reduce waste and energy use associated with their applications. [2]

This makes bioreactors crucial for current challenges in large-scale biological production. As a result, they are extensively utilized in the pharmaceutical, agri-food, and environmental

sectors. However, due to their applications, bioreactors are complex devices. This complexity is exacerbated by the fact that, in the pursuit of improving yields, bioreactors are becoming increasingly automated. Nevertheless, despite their complexity, bioreactors are also used in educational and innovative settings. For schools and startups, bioreactors offer exciting learning and product development opportunities. Schools can use them as educational tools to teach concepts of cell culture and biological production, while startups can leverage their flexibility to develop new biological products, from medical biotechnology to fermented food products.[3]

While automated systems offer unparalleled precision, real-time monitoring, and the ability to react to changes, they are prohibitively expensive for schools and startups. Despite this, we recognize the importance of incorporating vital parameters such as pH control into these systems, understanding the significance of it in biological processes. We have identified an opportunity to reduce the costs of automating bioreactors used in post-secondary education and startups. In this article, we explain how we have succeeded in significantly reducing these costs using coding software, a 3D printer, and affordable equipment.[3] Furthermore, through these explanations, we believe that our solution will convince you that cost-effective automation is the inevitable path for the future of bioreactors in schools and startups.

MARKET STUDY

I. GROWING INTEREST FOR BIOREACTORS IN BIOLOGY.

The global bioreactors market size was evaluated at USD 9.3 billion in 2022 and it is expected to hold around USD 30.42 billion by 2032, growing at a CAGR (Compound Annual Growth Rate) of 12.58% between 2023 and 2032.[4] In recent years, the global landscape of biotechnology and pharmaceutical industries has witnessed a paradigm shift, with an increasing emphasis on sustainable and efficient methods of production. Central to this transformative wave is the rising interest in bioreactors:



FIGURE 1: EVOLUTION IN THE NUMBER OF CITATIONS OF "BIOREACTOR" ON PUBMED FROM 1990 TO 2024

This growing demand could be explained by the inherent benefits of bioreactors in facilitating efficient and scalable production processes, positioning them as standards in the industry.

II. OUR POSITION IN THE MARKETS

Our bioreactor is designed for small scale (2 to 5 liters), strategically positioned between the initial phases of culturing in flasks and the scale-up to 10 liters. This intermediate scale serves as a crucial testing ground for experimentation, feasibility studies, and meticulous process optimization. Starting small enables researchers to navigate the critical early stages of development, evaluating risks and identifying challenges before committing to larger production scales.

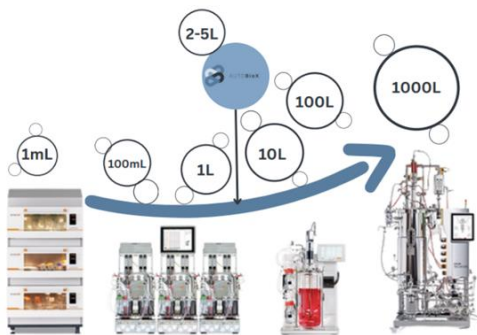


FIGURE 2: AUTOBIOX'S BIOREACTOR POSITION IN THE SCALE UP PROCESS

The flexibility of small-scale bioreactor systems facilitates adaptation to evolving project requirements, ensuring an agile response to emerging data. Basically, positioning within the 2 to 5-liter range establishes a strategic and adaptive framework, optimizing the bioprocess development journey for enhanced adoption by both big companies starting new projects and start-ups or academics with low budgets.

III. COMPETITORS

Our automated bioreactor, developed for its cost-effectiveness, and reliability, currently stands in a unique position within the market. There is currently no direct competitor that offers the same combination of automation, affordability, user-friendly and adaptability. This distinctive profile positions our bioreactor as a unique solution, meeting the demand for advanced technology without compromising on cost-effectiveness and reliability. In a landscape where specialized and advanced bioreactors are emerging, our product fills a crucial niche, setting it apart as an exceptional

choice for various bioprocessing applications. The only competitor that could position itself in the same market as us would be Jeulin, which develops small scale bioreactors for education. However, we do not consider Jeulin as a direct competitor, given that the bioreactor's capacity is significantly too small (20mL) and might not target researchers seeking to validate small-scale manufacturing processes. Finally, competitors or new entrants with a keen interest in DIY technological projects could indeed emerge. They might be capable of building a DIY bioreactor by purchasing individual components, although they would not leverage the Arduino technology developed by AutoBioX.

Bioreactors Comparison					
	USERFRIENDLY	AUTOMATION	CAPACITY	ADAPTABILITY	PRICE
Commercial		✓	✓	✓	1000€ - 5000€
AutoBioX	✓	✓	✓	✓	100€ - 150€
Jeulin	✓				46€

FIGURE 3: COMPARATIVE TABLE WITH BIOREACTORS FROM AUTOBIOX, JEULIN AND HOMEMADE BIOREACTORS

More globally in the world of bioreactors, big companies like Sartorius AG, Thermo Fisher Scientific, or Merck Group have been the main players for a long time. Depending on the options available, their prices range from 10,000 Eur to 50,000 Eur.

In the advanced therapies field, there has been a noticeable trend towards specialized instruments, with the development of bioreactors, fully automated and tailored for specific applications (such as those designed for cell and gene therapy).[5] This shift reflects a growing recognition of the requirements of automation and personalized bioreactors associated with advanced therapeutic areas. In this context, emerging contenders such as Miltenyi Biotec or Lonza have introduced a new era of bioreactors, characterized by advanced features and a higher price tag ranging from 100,000 to 200,000 Eur.

IV. A PROMISING TOOL FOR A MARKET-ENTRY

S (Strengths)	W (Weaknesses)	O (Opportunities)	T (Threats)
• Reduced manufacturing cost compared to commercial models available in the market • Possibility to customize the bioreactor (specific user needs) and Ability to adapt the bioreactor • Precise quality and performance control	• Limited by the bioreactor size • Reliability concerning certain probes • Bioreactor system safety (contamination, leaks, etc.)	• The demand for affordable solutions in biological research and biotechnology is steadily increasing • Possibility to introduce a unique solution in the market • Ability to customize the bioreactor according to customer needs, which can improve customer satisfaction and loyalty	• Growing competition from established commercial manufacturers in the market • The lack of technical knowledge and expertise can obstruct the bioreactor's implementation and usage for certain customers

FIGURE 4: SWOT ANALYSIS

After the SWOT assessment and due to its uniqueness, affordability, and adaptability to various projects, our bioreactor is a strong candidate for a market entry.

DESIGN AND IMPLEMENTATION: “CRAFTING THE FUTURE”

In a context where innovation is the key to progress, the possibilities of creating an innovative and accessible bioreactor are promising. Our project aims to address a crucial need in educational, startup, and small-scale applications by providing an automated bioreactor with pH monitoring capabilities. The high costs of current automated systems often exclude schools and new businesses from these essential technologies. However, by creating new approaches such as the use of affordable materials and open-source microcontrollers like Arduino, we have identified a unique opportunity to significantly reduce production costs while offering reliable automation. Our product positions itself as an affordable, user-friendly, and adaptable solution, filling a gap in the current market and facilitating access to advanced technologies for those exploring the fields of cell culture and biological production.

The development of our bioreactor will bring a solution that addresses the needs of diverse user groups. By defining the requirements based on the demand, our concept revolves around creating a bioreactor equipped with a pH probe and Arduino setup, with the flexibility to incorporate additional probes tailored to specific user requirements.

I. STRATEGIC DESIGN APPROACH.

To ensure flexibility and adaptability, our design focuses on a modular architecture. Our bioreactor has a specialized container with a lid featuring multiple openings, providing the flexibility to insert various types of probes, including the essential pH probe and an agitator. This thoughtful design allows users to customize the bioreactor according to their specific monitoring and experimentation needs.

The heart of our bioreactor is our Arduino microcontroller linked to a sophisticated electrical board. This board incorporates essential elements such as LEDs, potentiometers, and an LCD for displaying digital values. The inclusion of a dedicated pH meter motherboard connected to the Arduino card ensures precise and real-time pH monitoring. We selected these components to establish a robust foundation for our bioreactor's core functionalities.

Our microcontroller, programmed to monitor pH levels in real-time, serves as the brain of the automation process. The coding implemented in Arduino facilitates dynamic adjustments based on pH readings, offering users a responsive and adaptive system. This automation feature is crucial in maintaining optimal conditions for various biological processes within the bioreactor.

II. DOCUMENTATION AND USER GUIDANCE.

Recognizing the importance of transparency and user-friendliness, we will provide comprehensive documentation.

Users will receive detailed assembly instructions, operational guidelines, and maintenance protocols. Our user manuals will serve as step-by-step guides, making the setup and operation of the bioreactor straightforward for users of varying technical backgrounds.

III. A DETAILED LOOK AT THE BIOREACTOR SCHEMATIC: AUTOBIOX

Our prototype image of the bioreactor provides a visual representation of our innovative design. In this representation, strategically positioned probes, a dynamic agitator, and a bioreactor vessel are interconnected. Each component has been thoughtfully integrated to ensure optimal functionality, enabling precise and real-time monitoring of cell culture conditions. The installation is linked to an Arduino microcontroller. This image represents the technological approach we have taken to automate and regulate the bioreactor's parameters.



FIGURE 5: PROTOTYPE OF AUTOBIOX

And as we said, the cover will have a lid with a few different openings enabling the customer to choose the probes he wants to use and to incorporate them into his bioreactor. For a better understanding, here is a scheme that represents a multi opening lid.

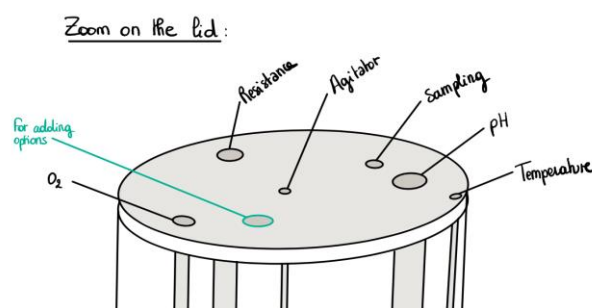


FIGURE 6: SCHEME OF THE MULTI-OPENING LID

IV. CONTINUOUS IMPROVEMENT

As we progress, we will establish a robust mechanism for collecting user feedback. This approach ensures that our bioreactor evolves in tandem with user needs and technological advancements. We are committed to follow a continuous improvement mindset.

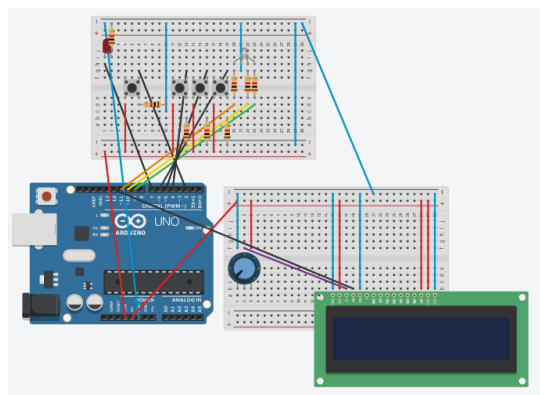


FIGURE 7: ELECTRICAL DIAGRAM OF THE ARDUINO ASSEMBLY

In conclusion, the feasibility of our project is not just a concept, it is a concrete plan to create a transformative solution to the bioprocessing industry. Here is a look at the electrical scheme of our Arduino monitoring system. We will discuss it in more detail in the third part.

AUTOMATION, PROGRAMMABILITY, AND ACCESSIBILITY: REVOLUTIONIZING BIOPROCESSING

I. AUTOMATION FOR ENHANCED EFFICIENCY

Our bioreactor is at the forefront of innovation, incorporating a high degree of automation to bioprocessing tasks. The integration of an Arduino microcontroller and a sophisticated electrical board allows automated control of essential parameters. Automation significantly reduces the manual workload, ensuring precision and consistency in the bioprocessing environment. Researchers and educators can focus on the scientific aspects of their work, confident that the bioreactor will handle routine monitoring and adjustments seamlessly.

The automated features of our bioreactor encompass a real-time pH monitoring, displayed on an LCD screen, presenting the numerical value of the solution's pH. Complementing this, an RGB LED adds a visual dimension to the monitoring process – illuminating in red when the pH falls below 6.8, green when it's between 6.8 and 7.2, and blue when it exceeds 7.2. This comprehensive system not only enhances operational efficiency but also serves as a fail-safe mechanism, reducing the potential for human error in pH control – a critical aspect in the delicate realm of bioprocessing.

Maintaining precise pH levels is paramount, as it directly impacts the success of biological processes within the bioreactor. The incorporation of visual indicators, coupled with real-time numerical feedback, provides a multi-sensory approach to pH monitoring, ensuring a swift and accurate response to deviations. This level of automation is particularly crucial in minimizing contamination risks associated with pH control, reinforcing the reliability and robustness of our

bioreactor in safeguarding the integrity of biological experiments and processes.

II. PROGRAMMABILITY FOR CUSTOMIZED EXPERIMENTS

One of the key advantages of our bioreactor lies in its programmability. Researchers, educators, and bioprocessing professionals can customize and program the bioreactor's operation based on the specific requirements of their experiments. The Arduino microcontroller, coupled with an intuitive programming interface, allows users to define and execute complex protocols with ease.

Whether it's adjusting temperature profiles, controlling nutrient dosing, or orchestrating sequential processes, the programmability of our bioreactor empowers users to conduct a wide range of experiments. This flexibility is particularly valuable in educational settings, where students can gain hands-on experience in designing and executing experiments in a controlled bioprocessing environment.

III. ACCESSIBILITY FOR DIVERSE USER GROUPS

Accessibility is a core principle guiding the development of our bioreactor. Traditional automated bioreactors often come with a high price tag, limiting their accessibility to well-established institutions or large corporations. In contrast, our bioreactor is designed to be cost-effective without compromising on functionality.

The use of open-source components, such as Arduino, not only contributes to affordability but also promotes accessibility by providing a platform that encourages tinkering, learning, and customization. The modular design further ensures that users with varying levels of technical expertise can assemble, operate, and maintain the bioreactor without significant barriers.

By making advanced bioprocessing technology accessible to educational institutions, startups, and small-scale applications, we aim to democratize the field and foster a new generation of innovators and researchers.

In summary, our bioreactor's automation, programmability, and accessibility converge to redefine the landscape of bioprocessing technology. It is not merely a piece of equipment; it is a tool that empowers users to push the boundaries of biological research and production, contributing to a more inclusive and dynamic biotechnology ecosystem.

DISCUSSION AND PERSPECTIVE

An evaluation of the accuracy of the pH probe compared to commercial probes will be necessary to more precisely assess the limits and prospects of our product. However, we can already discuss the identified limitations from our current

bioreactor and explore potential improvements to enhance its effectiveness and versatility in the future.

The current limitations of our product include the ability to measure only pH. Although pH is a crucial parameter, it would be wise to extend measurement capabilities to include other parameters such as temperature and oxygen concentration. This expansion would significantly increase the bioreactor's relevance across a broader range of applications, potentially reaching new audiences and increasing interest in our product. Currently, pH regulation requires manual adjustments of pH measurements. A crucial future perspective would be the implementation of an automatic pH regulation system, ensuring constant and precise control throughout the process and also mitigating potential human errors. Finally, the bioreactor is presently tailored for users seeking a volume ranging from 2 to 5 liters. This poses a constraint, limiting its adaptability across diverse scientific experimentation scales. If our target audience aims to upscale (increase volume), we currently lack a suitable solution due to our volume restriction capped at 5 liters.

Following the identification of these limitations, future prospects for our bioreactor include the integration of other measurement parameters, diversification of proposed volumes, greater customization of Arduino coding to meet specific user needs, and a move towards full automation. These developments would ensure a more extensive use of the bioreactor, reinforcing its adaptability and relevance in various research contexts. In conclusion, although the bioreactor currently has certain limitations, future improvement prospects position it as a promising solution for the diverse needs of biotechnology research.

CONCLUSION

In conclusion, AutoBioX represents a significant advancement in the field of automated bioreactors, blending technological innovation with operational practicality. From a scientific standpoint, our bioreactor, equipped with an integrated automated pH measurement system using an Arduino microcontroller, provides an advanced solution for cell culture and biological production. The advanced automation allows precise control of critical environmental parameters, ensuring optimal conditions for cell growth and the production of specific biomolecules. On the commercial front, AutoBioX occupies a unique position in the market. Unlike traditional expensive bioreactors, our solution is designed with a specific focus on cost-effectiveness without compromising quality and functionality. Targeting educational institutions and startups specifically, our bioreactor offers an affordable and accessible alternative to automation, thereby contributing to expanding the scope of bioprocessing technology in these sectors. The strategic

position of AutoBioX in the market is reinforced by its modular design, user-friendly interface, and the ability to adapt to culture volumes ranging from 2 to 5 liters. These features make it an ideal tool for the initial phases of cell culture, allowing researchers to test and optimize their processes effectively before scaling up to larger production volumes. The SWOT analysis reveals the relevance and competitiveness of AutoBioX in the market. By clearly identifying strengths, weaknesses, opportunities, and threats, our team can shape a robust business strategy that capitalizes on the unique advantages of our bioreactor. In terms of future prospects, AutoBioX has the potential to become a major player in the biotechnology sector. The integration of new features, such as automated measurement of additional parameters, expanding the range of volumes, and more advanced automation, can further broaden its appeal and utility in diverse research applications. In summary, AutoBioX represents a successful fusion of scientific innovation and business considerations, offering an automated bioreactor solution that addresses current market needs while paving the way for exciting future opportunities in the bioproduction and biotechnological research domain.

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We want to express our gratitude to SupBiotech, and everyone involved for helping us complete this project. A special thanks goes to Mr. Julien Grimaud, whose guidance and support have been crucial elements in the development of our research project. His insightful advice, expertise, and dedication to our academic success have greatly enriched our learning experience and we would also like to express our gratitude to Baptiste Favard for his valuable guidance and invaluable support throughout our project. His contribution has been a source of enrichment, providing an additional perspective that has greatly contributed to the success of our work.

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Chlorella vulgaris: A Natural and Effective Solution for the Depollution of Wastewater from the Dairy Industry

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Abstract - With greenhouse gas emissions on the rise, the dairy industry is getting more attention for its environmental impact, leading to a need for eco-friendly wastewater solutions. This study looks into using Chlorella vulgaris as a natural way to tackle phosphate pollution in dairy wastewater. Showing impressive abilities to absorb phosphates, Chlorella vulgaris provides a sustainable method for dealing with pollution. The research suggests that Chlorella vulgaris could be a natural, effective, and environmentally friendly solution, fitting with the dairy industry's goal of being environmentally responsible and sustainable.

Index Terms - biomass, microalgae, Chlorella vulgaris, phosphate sequestration, photobioreactor, white water, dairy industry.

INTRODUCTION

In recent years, the increasing levels of greenhouse gas emissions have worsened pollution issues, highlighting the seriousness of the ongoing environmental crisis. In a time where people globally are more aware of the imminent threats of climate change, industries are under increased pressure to adopt sustainable practices to reduce their environmental impact. In this context, the dairy industry, a key player in the French economy, is facing heightened environmental concerns. Strict environmental regulations require companies to carefully manage their wastewater, emphasizing the urgent need to find effective solutions to clean up the pollution. [1] At the core of the wastewater problem in the dairy sector is the widespread presence of phosphate, a compound well-known for its significant role in water pollution.

Phosphate, a primary pollutant inherent in wastewater, precipitates far-reaching ecological implications, contributing to the degradation of aquatic ecosystems and the decline in water quality. Adherence to rigorous regulations is imperative for the dairy industry to curtail its environmental impact and fulfill its ecological responsibilities. [2]

This article describes a helpful natural solution for dealing with the problem of phosphate pollution in dairy wastewater: Chlorella vulgaris. [2]

Chlorella vulgaris is a small green microalgae with impressive properties that allow it to absorb phosphate while it grows. This makes it an attractive and eco-friendly option for improving water pollution. Thriving in the rich and cloudy environment of dairy wastewater, Chlorella vulgaris not only has the potential for sustainable recycling but also offers a practical way to reduce pollution. [3]

Subsequent sections of this article delve into the distinctive capabilities of Chlorella vulgaris, supported by experimental data illustrating its capacity to absorb phosphate from dairy wastewater, juxtaposed with its potential for growth in an opaque environment. Our experiences delineate the progression of C. vulgaris concentration and phosphate concentration under sterile conditions. Through this research, our objective is to establish Chlorella vulgaris as a natural, effective, and environmentally conscientious solution, thereby contributing to the enhancement of depollution endeavors within the dairy industry.

MATERIAL & METHOD

TEST WATER COMPOSITIONS:

Chlorella free environment (condition A) 208.32 ml milk (Blédina infantile milk, Ref: 3041091466846) at 480 mg phosphate per 1,000 ml milk (Ci: 480mg/L), topped up with distilled water (MQ water, distilled ultra-pure, Sup'Biotech) (up to 1,000 ml).

White-water-free environment (condition B): 200ml microalgae and make up to 1000ml with distilled water.

Test environment (condition C): 208.32 ml of milk with 480 mg phosphate for 1,000 ml of Blédina growth milk 3 months

(Ci: 480mg/L); 200 ml of microalgae (Roscoff) and make up to 1,000 ml.

Wastewater preparation:

Wastewater preparation is carried out considering the phosphorus concentrations present in white water, as well as the proportions between milk and water, in a total volume of 2 L. This preparation is carried out with an estimated safety margin of excess. A wastewater mixture is prepared in such a way as to be distributed equitably between the different conditions, thus ensuring that a homogeneous context is maintained. In general, white water has phosphate concentrations ranging from 20 to 250 mg/L. The decision was taken to create a white-water medium with a phosphate concentration set at 100 mg/L.

In the preparation of the mixtures, the use of a common mixture is avoided. For the negative medium condition, a mixture is prepared directly with a fixed concentration of 100 mg/L phosphate. For the test condition, on the other hand, the introduction of microalgae results in a dilution of the phosphate concentration.

For the negative medium condition, milk containing 480 mg phosphate per 1,000 ml milk (Ci: 480 mg/L) is used. A volume of 208.32 ml of milk is withdrawn using a graduated cylinder and micropipettes (**Biohit**) ($Ci \cdot Vi = Cf \cdot Vf$, so $Vi = (1000 \text{ ml} \cdot 100 \text{ mg/L}) / 480 \text{ mg} = 208.32 \text{ ml}$) and made up to 1,000 ml with distilled water (i.e. 791.68 ml of distilled water). The solution is prepared in a 1000mL beaker (Ci: initial concentration, Vi= initial volume, Cf: final concentration, Vf: final volume).

For the test medium condition, milk containing 480 mg phosphate per 1,000 ml milk (Ci: 480 mg/L) is also used. A volume of 208.32 ml of milk is taken ($Ci \cdot Vi = Cf \cdot Vf$, so $Vi = (1000 \text{ ml} \cdot 100 \text{ mg/L}) / 480 \text{ mg} = 208.32 \text{ ml}$) and topped up with distilled water to 800 ml (i.e. 591.67 ml distilled water), for subsequent addition of 200 ml microalgae. The final volume is adjusted to 1,000 ml, ensuring a phosphate concentration like that of the negative medium condition. This procedure is implemented to maintain identical phosphate concentration conditions in our wastewater samples. The solution is prepared in a 1000mL beaker.

Preparation of white water:

In a sterilized graduated cylinder, use micropipettes to withdraw 208.32ml of milk and pour into an autoclaved 1L bottle.

To collect 208.32 mL of *Blédilait* milk, a two-step approach is taken. Initially, a volume of 208 mL of milk is taken from a graduated cylinder. Next, 0.32 mL of the same milk is extracted using a pipette.

The two samples are then combined and poured into a 1 L bottle. Perform the same operation in a second autoclaved 1L bottle. Each bottle is topped up to 1L with distilled water.

Cultivation of *Chlorella vulgaris* under different conditions:

3 different culture conditions: condition B (*Chlorella vulgaris* + BBM nutrient medium), condition A (white water) and a condition C (*Chlorella vulgaris* + BBM nutrient medium + white water).

For the condition C, 200 ml of *Chlorella* is added to the first bottle of white water prepared earlier. To do this, the autoclaved graduated cylinder is used to accurately measure 200 ml of the solution from the *Chlorella vulgaris* bag. As recommended by Bioteos, 20% of the volume of the *Chlorella vulgaris* pouch is added to the final culture medium.

For the condition B, in a second 1000 ml bottle, use the autoclaved graduated cylinder to add 200 ml of solution from the *Chlorella vulgaris* bag (according to Bioteos, 20% of the volume of the bag with *Chlorella vulgaris* should be added to the final culture medium). Still using the graduated cylinder, make up to the 1,000 ml mark with distilled water (i.e. 800 ml).

For condition A, simply recover the bottle previously prepared.

Triplicates: Dispense 300ml of each medium into 3 x 500ml beakers, using a funnel to pour to the 300mL mark. Then use the 5ml volumetric pipette to add 3ml of BBM 1% culture medium ONLY (its composition is detailed in *Table 1*) to each triplicate of condition B and condition C.

Compound name	Reference	Concentration (g/L)
NaNO3	Ref: 71755-250G	30
MgSO4	Ref: M5921-500G	9
NaCl	cupboard FR	3
K2HPO4	Ref: P3786-100G	9
KH2PO4	Bioteos	21
CaCl2	Ref: C3306-100G	3
H3BO3	Ref: B0394-100G	1,4
Na2 EDTA	Ref: E5134	9
FeSO4.7H2O	Ref: 215422-5G	0,6
ZnSO4.7H2O	Ref: 221376-5G	0,0106
MnCl2.4H2O	Ref: 221269-100G	1,7
MoO3	Ref: M0753-100G	0,00852
CuSO4.5H2O	Bioteos	0,0121
KOH	Bioteos	3,72

Table 1: BBM 1% medium composition table

ANALYSIS OF WHITE-WATER DE POLLUTION BY CHLORELLA VULGARIS

Phosphate determination:

Phosphate concentrations were determined using the HI713 kit (HANAHI713-25, **Hanna Instruments**). The calibration procedure for this kit is carried out directly using a solution of known phosphate concentration, guaranteeing the accuracy and reliability of subsequent measurements by adjusting the system to reference values.

Phosphate measurement was carried out using a dedicated photometer (HI713 Checker® HC phosphates, **Hanna Instruments**) included in the HI713 kit. This photometer, specifically designed to match the standard solution, was calibrated using the latter. In this way, the photometer enables direct determination of the phosphate concentration of the samples analyzed.

Samples are measured in duplicate when analyzed with the HI713 kit. A 10 mL sample of wastewater is taken, followed by the addition of a sachet of the kit's developer to the sample. After stirring for 2 minutes, the sample is incubated at room temperature for 3 minutes. The phosphate concentration is then measured using the photometer included in the HI713 kit. Note that due to a lack of reagents, measurements were limited to 1 each for positive and negative media.

The results are then compiled in an Excel file.

Growth monitoring using Malassez cell counting:

Malassez cell counting was selected as the method of enumeration. The analysis process involved a series of steps. Firstly, the cell (**DUTSHER**) is moistened to enable a coverslip to be attached, enabling the sample to be deposited in the counting chamber.

Next, a sample is taken using a micropipette, then carefully deposited in the Malassez cell for observation. Microalgae growth was assessed by examining the sample under a microscope (Sup'Biotech), using a x40 objective for adequate resolution. Microalgae in the field of view were then counted.

To count cells using the Malassez counting chamber, one counts within five squares of the chamber and calculates the average to determine the number of cells per unit volume (figure 1).

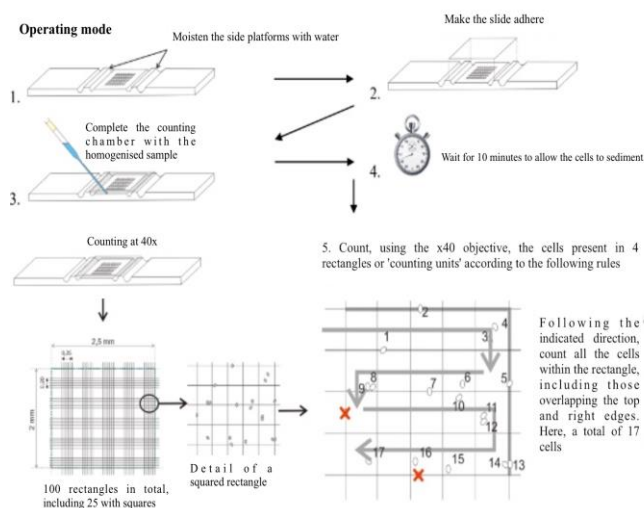


Figure 1: Operating mode to count with Malassez chamber.

Statistical analysis:

Statistical tests are used to demonstrate the significance of an experiment by assessing whether the observed differences between experimental groups are likely due to chance or if they are statistically significant.

To check whether differences between conditions are significant, statistical tests are carried out in two stages:

- Check whether the points follow a normal distribution: YES/NO
- If YES, then a Student test is performed
- If NO, a Wilcoxon test is performed.

RESULTS

I. EXPERIMENT REMINDER

This experiment, conducted under sterile conditions, aims to demonstrate the capability of the microalga *Chlorella vulgaris* in depolluting wastewater from dairy industries. Microalgae cultures are established in triplicates under three distinct conditions to ensure the reliability of experimental results and minimize experimental variations. The cultures are subjected to continuous agitation and either exposed or not exposed to light in a circadian cycle. The three experimental conditions are as follows:

- A: water + milk (= white wastewater)
- B: *Chlorella vulgaris* + water + nutrients
- C: *Chlorella vulgaris* + water + milk + nutrients.

The nutrients used are essential for the growth of the microalgae.

For measurements, samples are collected from each condition for the three triplicates to measure phosphate concentration using the Hannah instrument kit. Additionally, microalgal concentration over time is determined similarly, using a Malassez chamber and a microscope.

Measurements are taken every day, twice a day, between 8 am and 4 pm, over the course of a week.

The figure 2A presents a photo illustrating the setup of the experiment, in Sup'Biotech laboratory, conducted by seven engineering students.

II. CHLORELLA VULGARIS GROWS IN WHITE WATER.

Figure 3 provides a graphical representation of the evolution of *Chlorella vulgaris* concentration over time under two distinct conditions: the environment without the addition of white water (condition B) and the test environment (condition C).

Statistical analyses reveal a significant disparity between condition C and condition B from 33 hours to end, except for 48 hours and 103 hours, highlighted in figure 3 by a symbol (^). This significant difference suggests that the specific

conditions of condition C promote the growth of *Chlorella vulgaris*.



Figure 2A: Photography of the cultures under three different conditions, in triplicate, under agitation, in the presence of light, under sterile conditions.

Moreover, starting from 48 hours, a significant increase in the growth of *Chlorella Vulgaris* is observed in Condition B compared to the initial concentration, while in Condition C, no significant disparity is noted at different time intervals, as indicated by an asterisk (*). This observation underscores the positive influence of Condition B on the growth of *Chlorella Vulgaris*.

It is also noteworthy that the observed disparity in condition B persists beyond 48 hours except for 72 hours and 103 hours, suggesting that this condition remains conducive to the sustained growth of *Chlorella vulgaris*.

These results strengthen the argument in favor of the significant impact of environmental conditions on the growth dynamics of *Chlorella vulgaris*, highlighting important implications for understanding its behavior in specific contexts.

III- CHLORELLA VULGARIS PHOSPHATE 'S CONCENTRATION.

Figure 4 depicts the evolution of phosphate levels over time under three distinct conditions: the environment without *Chlorella* (condition A), the environment without white water (condition B), and the test environment (condition C). Statistical tests reveal significant differences between the initial ($t_{initial}$) and final (t_{final}) values in each condition, as indicated by an asterisk (*) in figure 3. In condition A, characterized by the exclusive presence of white water, a significant increase in phosphate levels is observed, likely due to contaminations in our environment, possibly emanating from microorganisms releasing phosphate. Regarding condition B, where the environment consists of water and *Chlorella*, a decrease in phosphate is observed due to the activity of *Chlorella* consuming phosphate. However, this decrease is less pronounced than in condition C, where a more substantial reduction in phosphate is observed. In condition

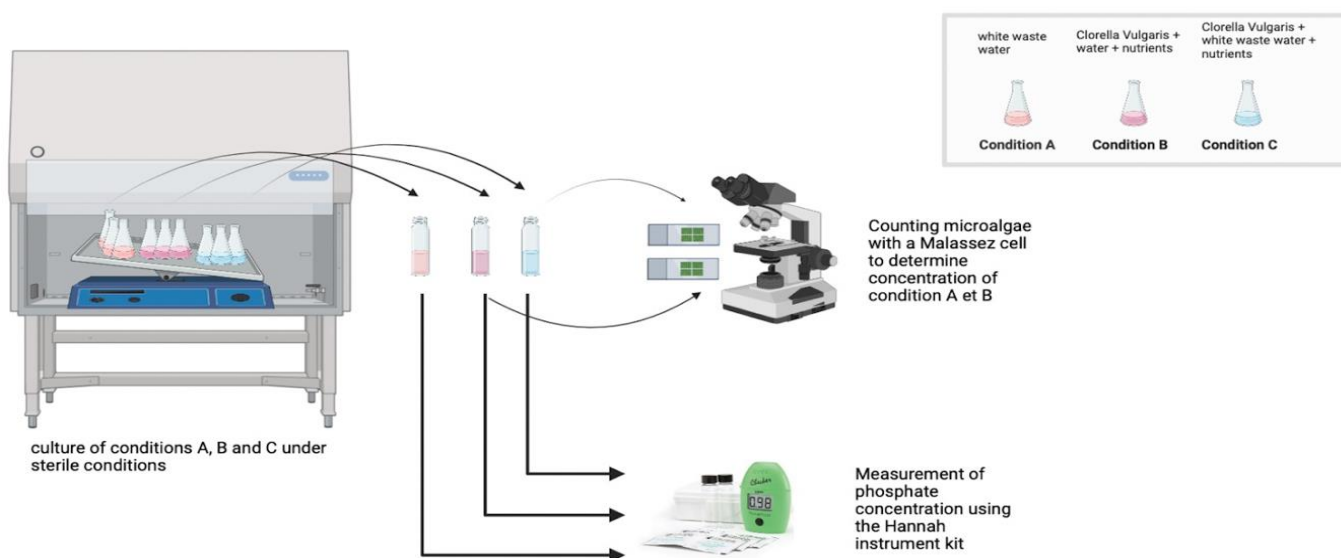


Figure 2B: Schematic diagram of the experiment: culture of the microalga *Chlorella vulgaris*, under three different conditions (A: white wastewater (pink), B: *Chlorella vulgaris* + water + nutrients (magenta), C: *Chlorella vulgaris* + white wastewater + nutrients (blue)), each condition in triplicate, with agitation, in the presence of light, under sterile conditions. On the figure, each color corresponds to a condition, repeated three times triplicate). To determine the concentration of conditions A and B, counting is carried out using a Malassez counting chamber. The phosphate concentration for each condition and each triplicate is realized thanks to the Hannah instrument kit.

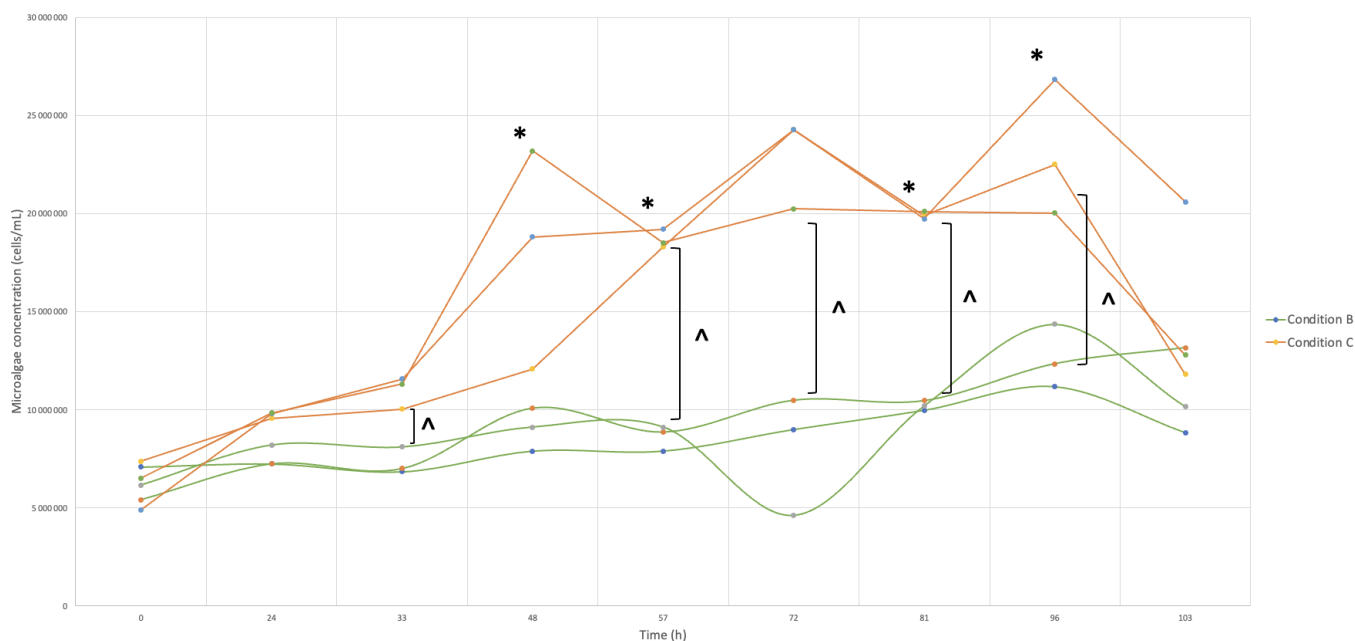


Figure 3: Evolution of *Chlorella vulgaris* concentration over time in two conditions (B and C). This figure illustrates the variations in *Chlorella vulgaris* concentration over time in two different conditions: the white-water-free environment (condition B) and the test environment (condition C). The stars (*) indicate a significant difference in a WSRT or Student test between the initial values (initial t) and each of the other values (from t=1 to final t) of each condition (B, and C). (*) indicates $p < 0.05$. The symbol (^) represents a significant difference in a WSRT or Student test between the values of the medium without white water and the test medium from initial t to final t. (^) indicates $p < 0.05$.

C, this significant difference indicates a substantial decrease in phosphate levels, suggesting a notable impact of the presence of *Chlorella vulgaris* on reducing phosphate levels in white water.

Furthermore, the results from figure 4 also demonstrate a disparity in the final phosphate concentrations between conditions A and C ($[A]PO > [C]PO$), as indicated by a (^). In condition B, where the absence of *Chlorella* is noted, the final phosphate concentration remains higher than that observed at the end of the experiment in the test condition (C). Since the same initial phosphate concentration was measured at the beginning of the experiment for these two conditions, it is evident that the test environment, in the presence of *Chlorella*, significantly influences the decrease in phosphate over time. These observations conclusively demonstrate that the presence of *Chlorella* is a determining factor in reducing phosphate levels. In environments initially equivalent in terms of phosphate concentration, a substantial decrease in phosphate is observed only when *Chlorella* is present in the environment.

These results underscore the importance of *Chlorella vulgaris* in the remediation of phosphate levels, suggesting its potential in breaking down this substance in aquatic environments.

DISCUSSION

The results presented herein, revealing the growth patterns of *Chlorella vulgaris* (C. vulgaris) and the temporal dynamics of

phosphate concentrations, were emanated from a unified set of experiments. As C. vulgaris undergoes proliferation, a significant decline in phosphate levels was observed, positing that the microalga utilizes phosphate as a vital component for its growth.

When examining the growth curves of *Chlorella*, it is notable that the microalga thrives in both white-water-free conditions (CB) and test environments enriched with white water (CC) (Figure 3). Notably, the growth in CC surpasses that in CB, indicative of the stimulatory effect of nutrients, particularly phosphate derived from white water, on *Chlorella*'s growth. However, post-t96, a reduction in biomass is noted, signifying cellular mortality.

Comparison between condition A (white-water *Chlorella*-free: CA) and the test environment revealed identical phosphate levels at the experiment's initiation (t0). Contrarily, by the conclusion of the experiment (t96) (Figure 4), a marked divergence in phosphate concentrations is remarkable. Specifically, our test environment demonstrated a significant reduction in phosphate levels, while CC exhibits an increase, implying potential contamination of our white-water *Chlorella*-free environments by biological constituents.

The promising results of our study suggest that the use of *Chlorella vulgaris* provides a sustainable method for combating phosphate pollution in dairy wastewater. However, several limitations need to be considered. Firstly, the issue of sterility may be raised, with the possibility of

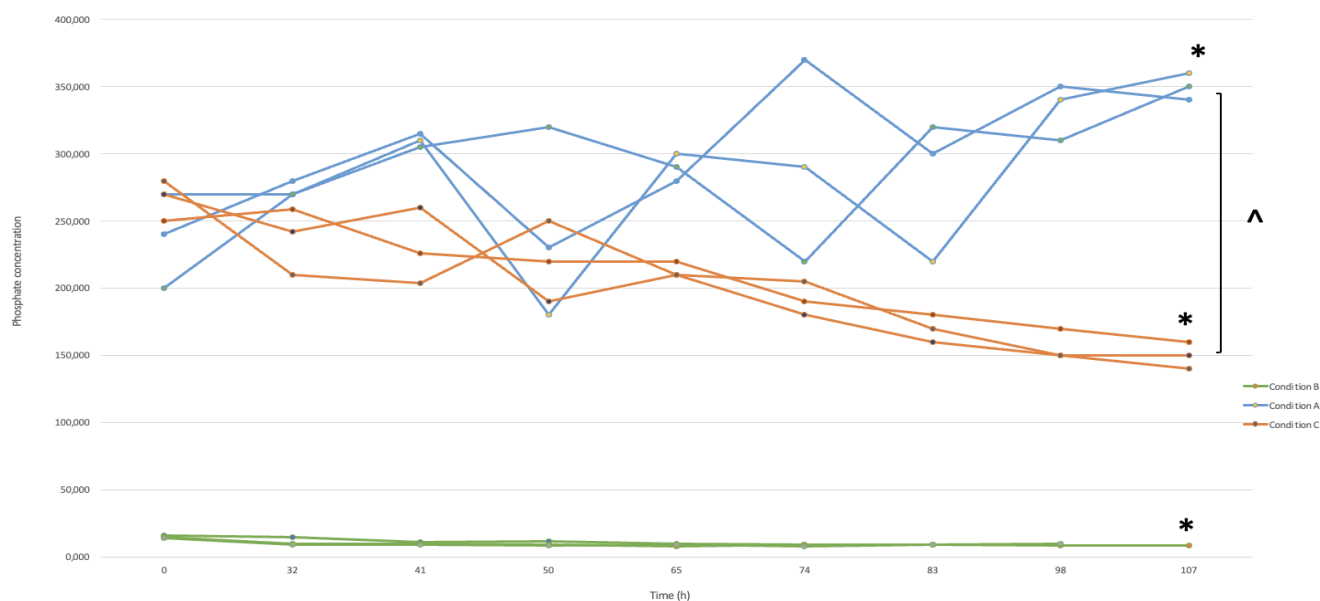


Figure 4: Phosphate evolution over time under different conditions. This figure illustrates the variations in phosphate levels over time in three different conditions: the *Chlorella*-free environment (condition A), the white-water-free environment (condition B), and the test environment (condition C). Asterisks (*) indicate a significant difference during a Student test between the initial ($t_{initial}$) and final (t_{final}) values of each condition (A, B, and C). (*) denotes $p < 0.05$.

contaminations, and verifying the initial purity of the microalgae would have strengthened confidence in the complete sterility of the conditions. Additionally, the use of the Hannah instrument kit for measuring phosphates may have implications for result accuracy, and adopting more precise instruments could have enhanced measurement reliability. Characterizing depollution solely through phosphate measurements may also be discussed, and adding a COD assessment would have provided a more comprehensive view of the process's effectiveness. A higher frequency of measurements would have brought a more detailed perspective on changes over time. Lastly, the reality of our innovation in sterile conditions may raise questions about its representativeness in real wastewater treatment environments. Future research should explore these limitations to refine the applicability of our solution in operational contexts.

Conclusively, our findings affirm the efficacy of employing *C. vulgaris* for the treatment of wastewater from milk industries. The microalga effectively mitigates phosphate levels, substantiating the viability of this technology.

CONCLUSION

In conclusion, the results of this experiment convincingly demonstrate the significant influence of environmental conditions on the dynamic growth of *Chlorella vulgaris* and its capacity to depollute wastewater from dairy industries. The experiment was conducted under sterile conditions with microalgae cultures established in three distinct experimental conditions. Daily measurements of phosphate concentration

and microalgal concentration revealed notable variations among the experimental conditions.

Condition C, characterized by the presence of *Chlorella vulgaris*, water, milk, and nutrients, exhibited significantly increased growth of *Chlorella vulgaris* compared to condition B, which included only *Chlorella vulgaris*, water, and nutrients. This difference persisted beyond 48 hours, indicating that the specific conditions of condition C favor sustained growth of *Chlorella vulgaris*.

Furthermore, analysis of phosphate levels revealed a significant decrease in condition C, highlighting the impact of the presence of *Chlorella vulgaris* on reducing phosphate levels in dairy wastewater. Comparisons with conditions A and B confirmed the influence of *Chlorella vulgaris* on phosphate degradation in white waters, emphasizing its essential role in regulating phosphorus levels.

In summary, these results strengthen the notion that *Chlorella vulgaris* can play a crucial role in the depollution of industrial wastewater, particularly from dairy industries. The implications of this research extend to understanding the growth dynamics of *Chlorella vulgaris* and its potential in phosphate degradation, paving the way for practical applications in sustainable water management. These findings offer promising perspectives for the development of innovative and sustainable environmental solutions based on the use of microalgae.

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